

Urinary Tract Obstruction

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KEY POINTS

- **Hydronephrosis and obstructive uropathy are not interchangeable terms—dilatation of the renal pelvis and calices can occur without obstruction, and urinary tract obstruction may occur in the absence of hydronephrosis.**
- **Urinary tract obstruction may be congenital or acquired. Acquired causes can be intrinsic or extrinsic.**
- **There is no single diagnostic technique that can safely diagnose obstruction. Therefore obstruction must still be considered in patients with worsening kidney function, chronic azotemia, or acute changes in kidney function or urine output, even in the absence of hydronephrosis.**
- **Both intrarenal and extrarenal factors combine to decrease glomerular filtration rate and renal blood flow during and immediately after release of obstruction.**
- **Urinary tract obstruction disrupts the ability to concentrate and dilute urine and impairs epithelial sodium, proton, and bicarbonate transport due to downregulation of most transporter proteins in the tubules.**
- **Urinary tract obstruction results in development of tubulointerstitial fibrosis with irreversible kidney injury due to a complex series of mechanisms involving multiple hormonal systems, as well as numerous cellular and molecular pathways.**

In adults, 1.5 to 2.0 L of urine flows daily from the renal papillae through the renal pelvis, ureter, bladder, and urethra in an uninterrupted, unidirectional flow. Any obstruction of urinary flow at any point along the urinary tract may cause retention of urine and increased retrograde hydrostatic pressure, leading to kidney damage and interference with waste and water excretion, as well as fluid and electrolyte homeostasis. Because the extent of recovery of kidney function in obstructive nephropathy is related inversely to the extent and duration of obstruction, prompt diagnosis and relief of obstruction are essential for effective management. Fortunately, urinary tract obstruction in most cases is a highly treatable form of kidney disease.

Several terms describe urinary tract obstruction, and definitions may vary.¹⁻³ In the following discussion,

“hydronephrosis” is defined as a dilatation of the renal pelvis and calyces proximal to the point of obstruction. “Obstructive uropathy” refers to blockage of urine flow due to a functional or structural derangement anywhere from the tip of the urethra back to the renal pelvis that increases pressure proximal to the site of obstruction. Obstructive uropathy may or may not result in renal parenchymal damage. Such functional or pathologic parenchymal damage is referred to as *obstructive nephropathy*. It should be noted that hydronephrosis and obstructive uropathy are not interchangeable terms—dilatation of the renal pelvis and calyces can occur without obstruction, and urinary tract obstruction may occur in the absence of hydronephrosis.

PREVALENCE AND INCIDENCE

The incidence of urinary tract obstruction varies widely among different populations and depends on concurrent medical conditions, sex, and age. Unfortunately, epidemiologic reports have been based on studies of selected “populations,” such as women with high-risk pregnancies and data from autopsy series. In the United States, it has been estimated that 166 patients per 100,000 population had a presumptive diagnosis of obstructive uropathy on admission to hospitals in 1985.⁴ The introduction of routine prenatal ultrasound (US) scanning resulted in an increasing number of infants suspected of having urinary tract obstruction,⁵ and with the increasing age of the population during the past 25 years, the incidence of obstructive uropathy may be expected to increase even more.

A review of 59,064 autopsies of persons varying in age from neonate to 80 years noted hydronephrosis as a finding in 3.1% of cases (3.3% in men and 2.9% in women). In children younger than the age of 10 years, representing 1.5% of all autopsies, the principal causes of urinary tract obstruction were ureteral or urethral strictures and neurologic abnormalities. It is unclear how frequently these abnormalities represented incidental findings, as opposed to being recognized clinically. Until the age of 20, there was no substantial sex difference in frequency of abnormalities (for details, see Chapter 71). Between the ages of 20 and 60 years, urinary tract obstruction was more frequent among women, mainly due to the effects of uterine cancer and pregnancy. After the age of 60 years, prostatic disease raised the frequency of urinary tract obstruction among men above that observed among women.⁶ In an autopsy series of 3172 children, urinary tract abnormalities were identified in 2.5%. Hydroureter and hydronephrosis were the most common findings, representing 35.9% of all cases.⁷ It is, however, not clear what proportion of such cases was diagnosed clinically before death.

Because a high proportion of these autopsy-detected cases of obstruction likely went undetected during life, the overall prevalence of urinary tract obstruction is likely far greater than reports suggest. This conclusion is reinforced by the fact that there are several common but temporary causes of obstruction, such as pregnancy and renal calculi.

CLASSIFICATION

Classification of urinary tract obstruction can be by duration (i.e., acute or chronic⁸), by whether it is congenital or acquired, and by its location (upper or lower urinary tract, supravescical, vesical, or infravescical, etc.). Acute obstruction may be associated with sudden onset of symptoms. Upper urinary tract [ureter or ureteropelvic junction (UPJ)] obstruction may present with renal colic. Lower tract (bladder or urethra) obstruction may present with dysfunctional voiding. By contrast, chronic urinary tract obstruction may develop insidiously and present with few or only minor symptoms, as well as with more general manifestations. For example, recurrent urinary tract infections, bladder calculi, and progressive chronic kidney disease may all result from chronic obstruction. Congenital causes of obstruction

arise from developmental abnormalities, whereas acquired lesions develop after birth, as a result of either disease processes or medical interventions.⁹

ETIOLOGY

Because congenital and acquired urinary tract obstructions differ to a great degree in cause and clinical course, they are described separately.

CONGENITAL CAUSES OF OBSTRUCTION

The advent of antenatal ultrasound in the late 1970s and its subsequent gradual but ultimately widespread implementation exposed a hitherto unrecognized high percentage of fetuses with congenital anomalies of the kidney and urinary tract (CAKUT) in 0.5% to 3% with dilatation of the urinary tract being most prevalent.¹⁰⁻¹² It thereafter became evident that not all dilatations in children were synonymous with obstruction, especially when comparing these numbers with the historic rates of surgical procedures for urinary tract obstruction in the preultrasound era.¹³⁻¹⁵ Differentiating between systems that are obstructed and those that are merely dilated is difficult because there is no gold standard diagnostic modality capable of interrogating the pathophysiology of an observed dilatation. In the absence of such a diagnostic test, clinicians rely on repetitive ultrasound examinations and studies such as renography to determine whether a dilated system needs surgical intervention or if active surveillance can continue. Typical indications for intervention include increasing renal anteroposterior pelvic diameter >30 to 50 mm on ultrasound or differential kidney function <40% on scintigraphy.^{14,15} Box 39.1 details the most common causes of congenital obstruction and dilatation of the urinary tract; the topic is comprehensively covered in Chapter 71.

ACQUIRED CAUSES OF OBSTRUCTION

INTRINSIC CAUSES

Acquired causes of obstruction may be intrinsic to the urinary tract (i.e., resulting from intraluminal or intramural processes) or may arise from extrinsic causes (Box 39.2).

Intrinsic intraluminal causes of obstruction may be intrarenal or extrarenal. Intrarenal causes arise from the formation of casts or crystals within the renal tubules. These include uric acid nephropathy,¹⁶ which usually results from the large uric acid load released when chemotherapeutic agents abruptly destroy large numbers of tumor cells in the treatment of patients with malignant hematopoietic neoplasms. Uric acid nephropathy may also occur in the setting of disseminated adenomatous carcinoma of the gastrointestinal tract.¹⁷ Drug-induced precipitations have been known to occur with the use of sulfonamides,¹⁸ acyclovir,¹⁹ indinavir,²⁰ and ciprofloxacin.²¹ Sulfadiazine, a relatively lipophilic sulfonamide, penetrates into the brain and has proven to be an excellent treatment for toxoplasmosis. However, the same lipophilicity makes the drug prone to formation of intrarenal crystals, which can lead to acute kidney injury when the drug is given in large doses.^{18,22} In multiple myeloma,²³ a common renal

Box 39.1 Congenital Causes of Urinary Tract Obstruction

Ureteropelvic Junction

Ureteropelvic junction obstruction
Proximal and middle ureter
Ureteral folds
Ureteral valves
Strictures
Benign fibroepithelial polyps
Retrocaval ureter

Distal Ureter

Ureterovesical junction obstruction
Vesicoureteral reflux
Prune-belly syndrome
Ureterocele

Bladder

Bladder diverticula
Neurologic conditions (e.g., spina bifida)

Urethra

Posterior urethral valves
Urethral diverticula
Anterior urethral valves
Urethral atresia
Labial fusion

complication is “myeloma kidney,” a condition also known as “myeloma cast nephropathy.” The renal lesions (casts) are directly related to the production of monoclonal immunoglobulin free light chains (FLCs), which coprecipitate with Tamm-Horsfall glycoprotein (THP) in the lumen of the distal nephron, obstructing tubular fluid flow.^{24,25} Promising experiments have identified the determinants of the molecular interaction between FLCs and THP, which permitted development of a peptide that demonstrated strong inhibitory capability in the binding of FLCs to THP in vitro.²⁵

Several intrinsic intraluminal, extrarenal, or intraureteral processes may also cause obstruction. Nephrolithiasis represents the most common cause of ureteral obstruction in younger men.²⁶ In the U.S. population, the prevalence of kidney stones in adults older than 20 years rose from 3.2% in 1980 to 10.1% in 2016.²⁷ The significance of this is also reflected by the large number of hospital admissions due to calculus of the kidney and ureters, amounting to 166,000 hospital stays in 2006.²⁸ Calcium oxalate stones occur most commonly. Obstruction caused by such stones occurs sporadically and tends to be acute and unilateral, usually without long-term decrements in kidney function. Of course, when a stone obstructs a solitary kidney, the result can be anuric or oliguric acute kidney injury. Less common types of stones, such as struvite (ammonium-magnesium-sulfate) and cysteine stones, more frequently cause significant kidney damage because these substances accumulate over time and often form staghorn calculi. Stones tend to lodge and obstruct urine flow at narrowings along the ureter including the UPJ, pelvic brim, and ureterovesical junction.

Other processes that cause ureteral obstruction include papillary necrosis, blood clots, and cystic inflammation. Papillary necrosis²⁹ may result from sickle cell disease or trait, amyloidosis,³⁰ acute pyelonephritis, exposure to nonsteroidal antiinflammatory drugs, or diabetes mellitus. Acute obstruction may even require surgical intervention.³¹ Blood clots secondary to a benign or malignant lesion of the urinary tract or cystic inflammation of the ureter (ureteritis cystica) can also lead to obstruction and hydronephrosis.³²

Intrinsic intramural processes that cause obstruction include dysfunctional voiding and acute or chronic urinary retention. Bladder storage of urine and micturition require complex interplay of spinal reflexes, midbrain, and cortical function. Neurologic dysfunction occurring in diabetes mellitus, multiple sclerosis, spinal cord injury, cerebrovascular disease, and Parkinson disease can result from upper motor neuron damage. Such lesions may disturb the intricate balance governing storage and micturition, leading to detrusor overactivity, a low compliance, high-pressure bladder, and voiding dysfunction as a result of detrusor sphincter dyscoordination leading to bladder emptying issues. The ensuing high bladder pressures are thus transmitted to the upper tracts by vesicoureteral reflux or obstruction results from the increased hydrostatic pressures needed to stretch the hypertrophic detrusor muscle.^{33,34}

Various drugs may cause intrinsic intramural obstruction by disrupting the normal function of the smooth muscle of the urinary tract. Anticholinergic agents³⁵ may interfere with bladder contraction, whereas levodopa³⁶ may mediate an α -adrenergic increase in urethral sphincter tone, resulting in increased bladder outlet resistance. In all circumstances when the bladder does not void normally, kidney damage may develop because of recurrent urinary tract infections and increased detrusor pressures.

Acquired anatomic abnormalities of the wall of the urinary tract include ureteral strictures and benign, as well as malignant, tumors of the urethra, bladder, ureter, or renal pelvis.³⁷ Ureteral strictures may develop due to stone impaction and as a complication of ureteral instrumentation or surgery.^{38,39} Furthermore, radiation therapy in children⁴⁰ and in adults⁴¹ treated for pelvic or lower abdominal cancers, such as cervical cancer, can result in ureteral fibrosis, strictures, and subsequent complications.

Infectious organisms may also produce intrinsic obstruction of the urinary tract. *Schistosoma haematobium* afflicts nearly 100 million people worldwide. Although active infection can be treated and obstructive uropathy may resolve, chronic schistosomiasis (bilharziasis) may develop in untreated cases, leading to irreversible ureteral or bladder fibrosis and obstruction.⁴² Of other infections, the incidence of genitourinary tuberculosis has remained constant over the 30 years, amounting to 3% to 5% of patients with tuberculosis.⁴³ Mycoses caused by *Candida albicans* or *Candida tropicalis* may also result in obstruction due to intraluminal obstruction (fungus ball) or invasion of the ureteral wall.⁴⁴

EXTRINSIC CAUSES

Acquired extrinsic urinary tract obstruction occurs in a wide variety of settings. The relatively high frequency of obstructive uropathy from processes in the female reproductive tract,

Box 39.2 Acquired Causes of Urinary Tract Obstruction**Intrinsic Processes****Intraluminal**

Intrarenal
 Uric acid nephropathy
 Sulfonamides
 Acyclovir
 Indinavir
 Multiple myeloma
 Intraureteral
 Nephrolithiasis
 Papillary necrosis
 Blood clots
 Fungus balls

Intramural

Functional
 Diseases
 Diabetes mellitus
 Multiple sclerosis
 Cerebrovascular disease
 Spinal cord injury
 Parkinson disease
 Drugs
 Anticholinergic agents
 Levodopa (α -adrenergic properties)
 Anatomic
 Ureteral strictures
 Schistosomiasis
 Tuberculosis
 Drugs (e.g., nonsteroidal antiinflammatory agents)
 Ureteral instrumentation
 Urethral strictures
 Benign or malignant tumors of the renal pelvis, ureter, bladder

Extrinsic Processes**Reproductive Tract**

Females
 Uterus
 Pregnancy
 Tumor (fibroids, endometrial or cervical cancer)
 Endometriosis
 Uterine prolapse
 Ureteral ligation (surgical)
 Ovary
 Tuboovarian abscess
 Tumor
 Cyst

Males
 Benign prostatic hyperplasia
 Prostate cancer

Malignant Neoplasms

Genitourinary tract
 Tumors of kidney, ureter, bladder, urethra
 Other sites
 Metastatic spread
 Direct extension

Gastrointestinal System

Crohn disease
 Appendicitis
 Diverticulosis
 Chronic pancreatitis with pseudocyst formation
 Acute pancreatitis

Vascular System

Arterial aneurysms
 Abdominal aortic aneurysm
 Iliac artery aneurysm
 Venous
 Ovarian vein thrombophlebitis
 Vasculitides
 Systemic lupus erythematosus
 Polyarteritis nodosa
 Henoch-Schönlein purpura

Retroperitoneal Processes

Fibrosis
 Idiopathic
 Drug induced
 Inflammatory
 Ascending lymphangitis of the lower extremities
 Chronic urinary tract infection
 Tuberculosis
 Sarcoidosis
 Iatrogenic (multiple abdominal surgical procedures)
 Enlarged retroperitoneal nodes
 Tumor invasion
 Tumor mass
 Hemorrhage
 Urinoma

Biologic Agents

Actinomycosis

such as pregnancy and pelvic neoplasms, results in higher rates of urinary tract obstruction in younger women than in younger men.^{45,46} The advent of routine abdominal and fetal ultrasonography in pregnant women has revealed that 50% to 90% of women entering their third trimester demonstrate some degree of dilatation of the collecting system.⁴⁵ The obstruction that is usually mild and subsides several weeks after delivery occurs at the level where the ureters cross the pelvic brim.⁴⁵ Mechanical and hormonal factors affect these changes, in that hypertrophy of the Waldeyer sheath (the smooth muscle layer encompassing the distal ureter) and compression from the gravid uterus can impinge on the ureters. Additionally, high progesterone

levels reduce ureteral tone and decrease peristalsis.⁴⁵ As the uterus is normally dextro-rotated and the right ovarian vein crosses the ureter, dilatation is more common on the right side.^{45,47} Clinically significant obstructive uropathy in pregnancy almost always presents with flank pain.⁴⁸ In these cases, ultrasonography serves as a useful initial screening test⁴⁹ and magnetic resonance imaging (MRI) can be used if ultrasound is not conclusive.⁵⁰ In cases of symptomatic or significant hydronephrosis, relief of obstruction can be attained by placing an internal stent or percutaneous nephrostomy catheter.⁵¹ Clinically significant ureteral obstruction is rare in pregnancy, and bilateral obstruction leading to acute kidney injury is exceptionally rare.⁴⁸

Conditions in pregnancy that may predispose to obstructive uropathy and acute kidney injury include multiple fetuses, polyhydramnios, an incarcerated gravid uterus, or a solitary kidney.⁵²

Pelvic malignancies, especially cervical adenocarcinomas, represent the second most common cause of extrinsic obstructive uropathy in women.⁵³ In older women, pelvic organ prolapse leads mainly to lower urinary tract obstruction; however, approximately 7.7% of women scheduled for prolapse surgery had hydronephrosis on preoperative ultrasound.^{54,55} Other benign pelvic abnormalities may cause ureter obstruction including uterine fibroids and ovarian cysts.^{56,57} Pelvic lipomatosis, a disease with an unclear etiology seen more often in men, is another less common reason for compressive urinary tract obstruction.⁵⁸

Endometriosis affects the urinary tract in up to 5% of women afflicted, especially in cases of deep infiltrating endometriosis. The ureters are involved in 9% to 23% of cases, usually unilaterally and with a predilection for the left side.⁵⁹ Obstruction may arise due to the inflammation and adhesions caused by endometrial peritoneal deposits, leading to extrinsic compression of the ureters. An alternative theory contends that hematogenic spread may explain intrinsic ureteral involvement.⁶⁰ The onset of obstruction may be insidious, and the process is usually confined to the pelvic portion of the ureter. It is thus imperative to screen for obstruction in cases of advanced endometriosis. Ultrasound should be the first-line imaging modality; computed tomography (CT) scanning or MRI can help to determine more precisely the type of ureteral involvement.⁵⁹ It is of note that approximately 50% of inadvertent ligations of the ureter in abdominal and retroperitoneal operations occur during gynecologic and obstetric procedures.⁶¹

After the age of 60 years, obstructive uropathy occurs more commonly in men. Benign prostatic hyperplasia, which is by far the most common cause of urinary tract obstruction in men, produces symptoms of bladder outlet obstruction in approximately 75% of men aged 50 years and older.^{62,63} Upper urinary tract obstruction may also ensue in severe cases.⁶⁴ Presenting symptoms of bladder outlet obstruction include difficulty initiating micturition, weak urinary stream, dribbling at the end of micturition, incomplete bladder emptying, and nocturia. The diagnosis may be established by history and urodynamic studies, as well as imaging in some cases.^{62,63,65}

Urologic malignancies have, in light of their anatomic origin, a particular propensity to cause both upper and lower urinary tract obstruction. Urothelial carcinoma, which accounts for 3% of global cancer diagnoses,^{66,67} can manifest anywhere along the lining of the urinary tract and may lead to obstruction at the ureterovesical junction in cases of bladder cancer or anywhere along the ureter. Presence of hydronephrosis in bladder cancer and upper urinary tract urothelial carcinoma has been found to adversely affect overall prognosis.^{68,69} Despite stage migration to more organ-confined disease in the era of prostate-specific antigen, obstruction due to prostate cancer compressing the bladder neck and invading the ureteral orifices is still relatively common.^{70,71} Urinary tract obstruction in advanced and metastatic prostate cancer can have a varied presentation because it may occur in multiple

anatomic locations including the ureter and pelvic lymph nodes.⁷¹

Gastrointestinal processes may also cause obstructive uropathy. Inflammatory bowel disease (IBD) may extend into the retroperitoneum, leading to obstruction of the ureters.⁷² Additionally, the malabsorption associated with IBD and bariatric surgery can lead to nephrolithiasis due to metabolic abnormalities including hyperoxaluria and hypocitraturia.^{73,74} Disorders of the colon including diverticulitis and fecal impaction may also on occasion lead to ureteral obstruction.^{75,76} Acute and chronic pancreatitis are also rare causes of mainly right-sided hydronephrosis.⁷⁷

Vascular disease also constitutes a rare cause of obstruction. Aneurysms of the abdominal aorta and iliac arteries may lead to ureteral obstruction either by direct pressure of the aneurysm or as a result of the associated retroperitoneal fibrosis.^{78,79} In addition, and also rarely, vasculitis caused by systemic lupus erythematosus,⁸⁰ polyarteritis nodosa,⁸¹ granulomatosis with polyangiitis,⁸² and Henoch-Schönlein purpura^{83,84} has been reported to cause obstruction.

Retroperitoneal neoplastic processes can obstruct the ureters at multiple levels by direct extension, encasement, or invasion as seen with periaortic nodal masses resulting from primary nodal disease, such as lymphoma or retroperitoneal metastases from urologic, gastrointestinal, or germ cell tumors.⁸⁵ Relief of obstruction can be achieved by stenting or nephrostomy tube placement.⁸⁵ Idiopathic, retroperitoneal fibrosis, on the other hand,^{86,87} usually involves the middle third of the ureter and affects men and women equally, predominantly those in the fifth and sixth decades of life.⁸⁷ Retroperitoneal fibrosis may also be drug induced (e.g., methysergide), or it may occur as a consequence of scarring from multiple abdominal surgical procedures.⁸⁷ It may also be associated with conditions as varied as gonorrhea, sarcoidosis, chronic urinary tract infections, Henoch-Schönlein purpura, tuberculosis, biliary tract disease, and inflammatory processes of the lower extremities with ascending lymphangitis.⁸⁷

Rare childhood tumors such as pelvic neurofibromas can induce upper urinary tract obstruction in up to 60% of patients.⁸⁸ Wilms tumor may obstruct via local compression of the renal pelvis,⁸⁹ and neuroblastoma may cause both upper and lower urinary tract obstruction.⁹⁰ Miscellaneous inflammatory processes can also result in obstruction. These include granulomatous causes such as sarcoidosis⁹¹ and chronic granulomatous disease of childhood.⁹² Amyloid deposits may produce isolated involvement of the ureter. Furthermore, a pelvic mass or inflammatory process associated with actinomycosis may cause external ureteral compression.^{93,94} Retrovesical echinococcal cysts can also impede urine flow.⁹⁵ Retroperitoneal malacoplakia can also be a rare cause of urinary obstruction.⁹⁶

CLINICAL ASPECTS

Urinary tract obstruction may cause symptoms referable to the urinary tract. However, even patients with severe obstruction may be asymptomatic, especially in settings where the obstruction develops gradually, or in patients with spinal cord injury.⁹⁷ The clinical presentation often depends on the rate of onset of the obstruction (acute or

chronic), the degree of obstruction (partial or complete), whether the obstruction is unilateral or bilateral, and whether the obstruction is intrinsic or extrinsic. Pain in obstructive uropathy is usually associated with obstruction of sudden onset, as seen with kidney stones, blood clots, or sloughed papillae, and appears to result from abrupt stretching of the renal capsule or the wall of the collecting system, where C-type sensory fibers are located. The severity of the pain appears to correlate with the rate, rather than the degree, of distention. The pain may present as typical renal colic (sharp pain that may radiate toward the groin), or, in patients with reflux, the pain is primarily in the flank during micturition. With UPJ obstruction, flank pain may develop or worsen when the patient consumes large quantities of fluids or receives diuretics.⁹⁸ Early satiety and weight loss may be another symptom.⁹⁹ It can be difficult to differentiate symptoms of obstruction from those of gastrointestinal disease.

Sometimes, patients notice changes in urine output as obstruction sets in. Bladder outlet obstruction may cause urinary retention, whereas anuria can occur due to bilateral ureteral obstruction or obstruction of a solitary kidney at any level. Usually, however, obstruction occurs with no change in urine output. Recurrent urinary tract infections may be the only sign of obstruction, particularly in children. Bladder outlet obstruction as seen in prostatic and urethral stricture disease often presents with difficulty initiating micturition, decreased size or force of the urinary stream, postvoid dribbling, and incomplete emptying.^{100,101} Irritative bladder symptoms such as frequency, urgency, and dysuria may result from urinary tract infection. Appearance of obstructive symptoms synchronous with the menstrual cycle may also be a sign of endometriosis.¹⁰²

On physical examination, several signs may suggest urinary obstruction. A palpable abdominal mass, especially in neonates, may represent hydronephrosis and in all age groups, a palpable suprapubic mass may represent a distended bladder.¹⁰³ On laboratory examination, proteinuria, if present, is generally less than 2 g/day. Microscopic and gross hematuria is common findings in the setting of nephrolithiasis, urologic malignancies, and when urinary tract infection complicates obstruction.¹⁰⁴ The urine sediment is often unremarkable. Less common manifestations of urinary tract obstruction include deterioration of renal function without apparent cause, hypertension,¹⁰⁵ polycythemia, and abnormal urine acidification and concentration capacity.

DIAGNOSIS

Careful history and physical examination represent the cornerstone of diagnosis, often leading to detection of urinary tract obstruction and suggesting the reason for it. Although challenging, history and physical examination should focus on evaluation so that the minimum amount of time and expense are incurred in determining the cause of the obstruction.

HISTORY AND PHYSICAL EXAMINATION

Important information in the history includes the type and duration of symptoms (voiding difficulties, flank pain,

urinary retention, and decreased urine output); presence or absence of urinary tract infections and their number and frequency (especially in children); pattern of fluid intake; and any symptoms of advanced chronic kidney disease (such as fatigue, sleep disturbance, loss of appetite, pruritis). In addition, relevant medical history should be reviewed in detail, looking for predisposing causes including stone disease, malignancies, gynecologic diseases, history of recent surgery, and drug use.

The physical examination should focus first on vital signs, which may provide evidence of infection (fever, tachycardia) or volume overload (hypertension). Evaluation of the patient's volume status will guide fluid therapy. The abdominal examination may reveal a flank mass, which may represent hydronephrosis (especially in children), or a suprapubic mass, which may represent a distended bladder. Features of chronic renal failure, such as pallor (anemia), drowsiness (uremia), neuromuscular irritability (metabolic abnormalities), or pericardial friction rub (uremic pericarditis), may also be noted. A thorough pelvic examination in women and a rectal examination for all patients are mandatory. A careful history and a well-directed and complete physical examination often reveal the specific cause of urinary obstruction. Coexistence of upper urinary tract obstruction and infection is a urologic emergency, and appropriate studies (US, CT, MRI) must be performed promptly so that the obstruction can be characterized and relieved.

BIOMARKERS FOR EVALUATION OF URINARY TRACT OBSTRUCTION

The effects of chronic obstruction on kidney function are the result of a complex series of events that profoundly and progressively alter all components of glomerular and tubular functions. The consequence is the development of tubulointerstitial injury characterized by dynamic changes involving tubular atrophy, inflammatory cell infiltration, and interstitial fibrosis. This leads to irreversible loss of kidney function and obstructive nephropathy, which may even continue to progress after relief of obstruction. Because the complexity of this process involves almost every cell in the kidney and many pathways, there have been many attempts to identify biomarkers, which predict the course of obstruction. These biomarkers are characterized by the current understanding of the pathophysiology of urinary tract obstruction on kidney injury and currently include biochemical assays of blood and urine, as well as many imaging tools as detailed later.

BIOCHEMICAL EVALUATION OF BLOOD AND URINE

The laboratory evaluation includes urinalysis and examination of the sediment on a fresh specimen by an experienced observer. Unexplained renal failure with benign urinary sediment should suggest urinary tract obstruction. Microscopic hematuria without proteinuria may suggest calculus or tumor. Pyuria and bacteriuria may indicate pyelonephritis; bacteriuria alone may suggest stasis. Crystals in a freshly voided specimen should lead to consideration of nephrolithiasis or intrarenal crystal deposition.

Hematologic evaluation includes the hemoglobin/hematocrit and mean corpuscular volume (to identify anemia of chronic renal disease) and white blood cell count (to identify possible hematopoietic system neoplasm or infection). Serum electrolytes (Na^+ , Cl^- , K^+ , and HCO_3^-), blood urea nitrogen concentration, creatinine, Ca^{2+} , phosphorus, Mg^{2+} , uric acid, and albumin levels should be measured. These will help identify disorders of distal nephron function (impaired acid excretion or osmoregulation) and uremia. Urinary chemistries may also suggest distal tubular dysfunction (high urine pH, isosthenuric urine) and inability to reabsorb sodium normally (urinary Na^+ >20 mEq/L, fractional excretion of Na^+ [FENa] $>1\%$, and osmolality <350 mOsm/L). Alternatively, in acute obstruction, urinary chemistry values may be consistent with prerenal azotemia (urinary Na^+ <20 mEq/L, FENa $<1\%$, and osmolality >500 mOsm/L).⁸

Novel biomarkers relevant for functional and cellular and molecular changes are being developed as an index of kidney injury and to predict functional reserve or recovery after reconstruction. Simple tests examining the value of risk factor proteins have suggested that elevated levels of neutrophil gelatinase-associated lipocalin and β_2 -microglobulin are present in the urine from obstructed kidneys.¹⁰⁶ Attempts to predict the clinical outcome of congenital unilateral UPJ obstruction in newborns by urine proteome analysis reveal an example of this powerful new technology. Polypeptides in the urine were identified and enabled diagnosis of the severity of obstruction, and using this technique the clinical evolution was predicted with 94% precision in neonates,¹⁰⁷ whereas the precision was only 20% in older children with UPJ obstruction.¹⁰⁸ Thus far, application of large-scale urinary proteomic analysis holds promise for better classification of individuals with hydronephrosis for early selection of surgical candidates,¹⁰⁹ but long-term follow-up studies are warranted to determine the true clinical value of this diagnostic approach.¹¹⁰ Use of a proteomic analysis of fetal urine using a 12-peptide signature expressed in fetuses who go on to develop end-stage renal disease by the age of 2 years has recently shown promise in determining prognosis of posterior urethral valves.¹¹¹

In an attempt toward improved understanding of obstructive nephropathy and improved translatability of the results to clinical practice, a systems biology approach combining omics data of both human and mouse obstructive nephropathy was developed. Using the urinary miRNome of infants with UPJ obstruction and the kidney tissue miRNome and transcriptome of the corresponding neonatal partial unilateral ureteral obstruction (PUUO) mouse model revealed that let-7a and miR-29b are potentially involved in the development of fibrosis in UPJ obstruction via the control of DTX4 in both humans and mice.¹¹²

EVALUATION BY MEDICAL IMAGING

The history, physical examination, and initial laboratory studies should guide the medical imaging evaluation. Pain, degree of impaired kidney function, and the presence of infection dictate the speed and nature of the evaluation. Numerous imaging techniques are available; each has its own advantages and disadvantages, including the ability to identify the site and cause of the obstruction and to

separate functional obstruction from mere dilatation of the urinary tract. Patient-specific factors, such as the risks of radiocontrast in the setting of impaired kidney function or exposure to radiation in pregnant women, must also be weighed.¹¹³

ULTRASONOGRAPHY

Ultrasonography is usually the initial investigative modality employed when urinary tract obstruction is suspected. Ultrasound provides excellent anatomic information about obstruction, primarily a dilated collecting system and the degree hereof. It can also provide information regarding the level and cause of obstruction. Ultrasound is safe and readily available and avoids ionizing radiation, thus making it ideal for pregnant¹¹³ and pediatric patients.¹¹⁴ Moreover, as no radiographic contrast is required, it is well suited in the assessment of suspected obstruction in the setting of impaired kidney function, where iodinated contrast administration should be used cautiously¹¹⁵ and in patients allergic to contrast material.¹¹⁶ It may also determine the size and shape of the kidney and may demonstrate thinned cortex in case of severe long-standing hydronephrosis (Fig. 39.1). Additionally, ultrasound may detect perinephric fluid collections, which may indicate urine leakage from an obstructed system or signs of perinephric abscess.¹¹⁷

Although ultrasound has a sensitivity of 98% for detecting hydronephrosis, it does not provide physiologic or functional data with respect to the dilatation observed, nor

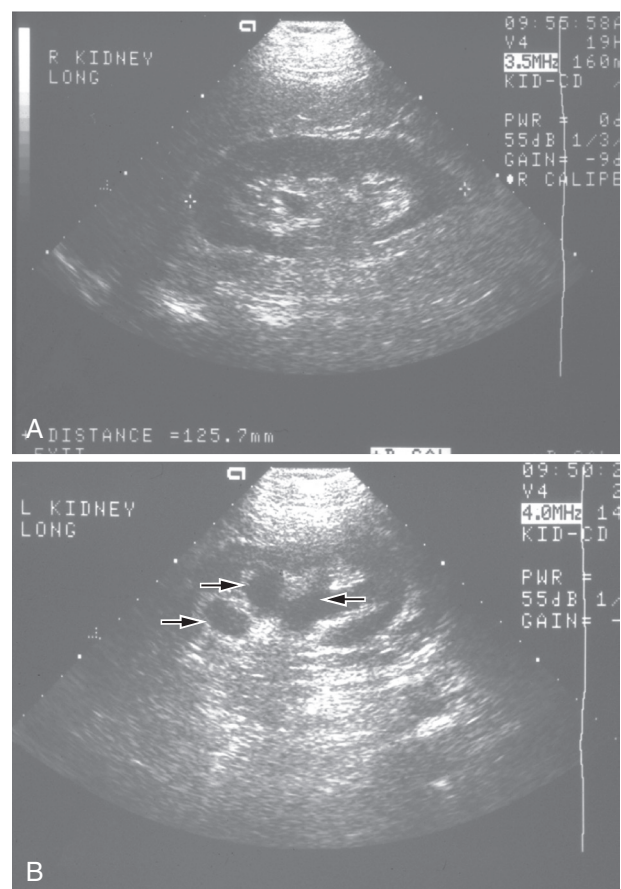


Fig. 39.1 Renal ultrasound. (A) Normal kidney. (B) Hydronephrotic kidney: dilated calices and pelvis (arrows).

does ultrasound provide information on kidney function. The specificity of ultrasound has been criticized, as there are multiple conditions other than obstruction that can cause dilatation, such as vesicoureteral reflux, a dilated extrarenal pelvis, and residual dilatation after previous obstruction, which can lead to false-positive interpretations.¹¹⁴ False-negative results are also known to occur, especially during the first 48 hours of obstruction,^{114,118-120} in cases of dehydration, staghorn calculi, nephrocalcinosis,¹²⁰ and retroperitoneal fibrosis,¹²¹ as well as with tumor encasement of the collecting system.¹²² A dilated collecting system without obstruction may also be observed in up to 50% of patients with urinary diversion through ileal conduits, especially on the left side.¹²³ In suspected stone disease, ultrasound is recommended as the initial investigative imaging modality in young patients and pregnant women; however, for adults with renal colic and especially those who are obese, most guidelines recommend the use of CT because its accuracy for diagnosing and characterizing stones, location, and burden exceeds that of ultrasound.¹²⁴

DUPLEX DOPPLER ULTRASONOGRAPHY

To increase ultrasound specificity toward differentiating between obstructed and nonobstructed dilatations, attempts have been made to harness the physiologic information emanating from the intrarenal vascular dynamics associated with an obstructive state by employing Doppler ultrasound. Doppler's ability to quantify intrarenal flow rates has therefore been used in calculating a resistive index, a surrogate measure for intrarenal vascular resistance. Studies have shown that a resistive index above the upper limit of 0.7 is associated with significant obstruction.¹²⁵ Diuretic challenge to the kidney and comparison of the resistive index of the two kidneys has also been shown to further enhance diagnostic accuracy.^{125,126} Despite early promising results, resistive index diagnostics have not been widely adopted, as outcomes have been inconsistent in pediatric patients and the elderly, as well as in cases of partial, intermittent, and acute obstructions.¹²⁵

INTRAVENOUS UROGRAPHY

Intravenous urography (also known as “intravenous pyelography”) was for many years the imaging modality of choice for patients suspected of having urinary tract obstruction (Fig. 39.2). Intravenous urography requires administration of iodinated contrast, is time consuming to perform, and for all purposes, it is obsolete in institutions where CT is available.

COMPUTED TOMOGRAPHY

CT as an imaging modality has become ubiquitous in urologic diagnostics, especially in urothelial disease, renal masses, work-up of hematuria, and obstructions.¹²⁷ CT has a particular advantage because it can visualize a dilated collecting system without the requirement for contrast enhancement (Fig. 39.3). Because of its exquisite sensitivity to density, CT can identify even radiolucent stones because uric acid stone density is at least 100 Hounsfield units (HU), which is higher than that of soft tissue (usually 10–70 HU).

CT is especially effective in identifying extrinsic causes of obstruction (e.g., retroperitoneal fibrosis, lymphadenopathy, and hematoma) and when contrast-enhanced CT is used,



Fig. 39.2 Intravenous pyelography. Normal right kidney and dilated collecting system on the left. The obstruction was relieved with a stent.

obstructing urothelial lesions in the ureters or bladder along with prostate pathology (Fig. 39.4) can easily be visualized.¹²⁷ Helical multiphasic contrast-enhanced CT has also proven to be an accurate and noninvasive method of demonstrating crossing vessels in UPJ obstruction.¹²⁸

Furthermore, CT can detect extra-urinary pathology and can establish non-urogenital causes of pain. All of these advantages establish non-contrast-enhanced helical CT as the diagnostic study of choice for the evaluation of the patient with acute flank pain.¹²⁹ Ultrasound may be the first method of diagnosis in this setting (Fig. 39.5), but CT resolution and depiction of details are usually superior to those of ultrasound.¹³⁰

In this regard, it must be emphasized that there is a minimal cancer risk from radiation exposure in connection with CT scans. This incurred theoretical risk must be considered in the context of the clinical benefit derived from a medically indicated CT and the likelihood of cancer occurrence in the general population. The optimal method of mitigating this risk is avoidance of unnecessary scans and optimizing the technical aspects of medically justified ones.¹³¹ This issue is especially pertinent in children who may be at higher risk of developing leukemias and brain tumors.¹³²

ISOTOPIC RENOGRAPHY

Isotopic renography and scintigraphy are helpful in diagnosing upper urinary tract obstruction and providing information on the differential renal function of both kidneys while avoiding the risk of radiocontrast agents.^{133,134} Radioisotope is injected intravenously, and its dynamic uptake and excretion by the kidneys are visualized and measured with a gamma camera. Although this

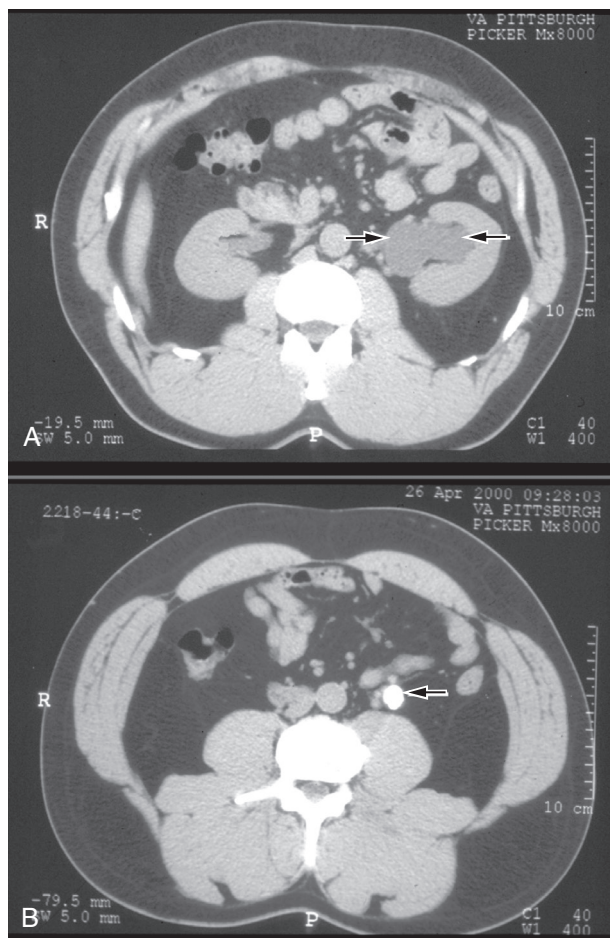


Fig. 39.3 Computed tomography, noncontrast study. (A) Left hydronephrosis: dilated renal pelvis (arrows), with normal kidney on the right. (B) Reason for obstruction: left midureteral stone (arrow).

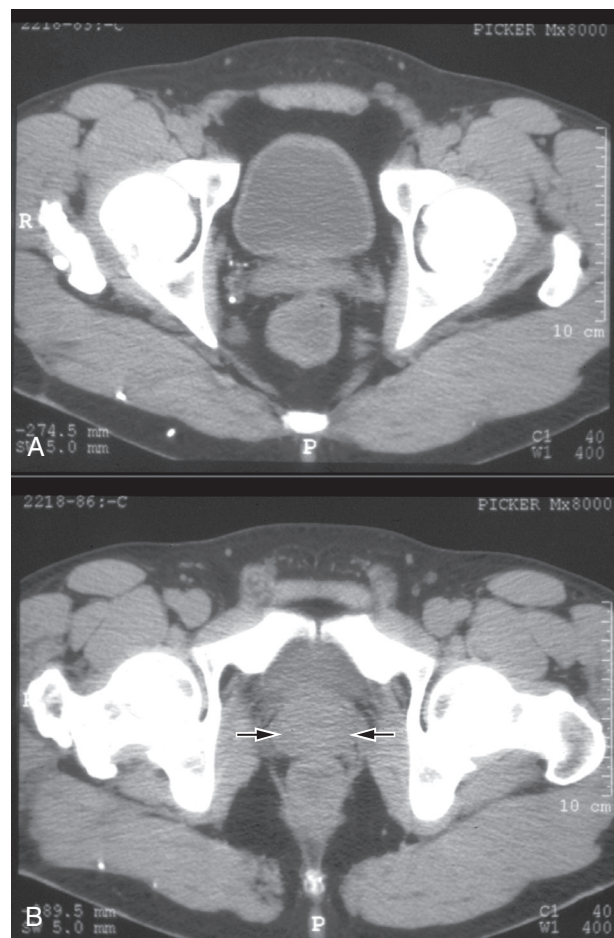


Fig. 39.4 Computed tomography of the pelvis. (A) Large post-voiding residual urine in the bladder. (B) Enlarged prostate (arrows), leading to urinary retention.

method gives a functional assessment of the obstructed kidney, anatomic definition is suboptimal compared with CT. Isotopic renography is typically used to estimate the fractional contribution of each kidney to overall kidney function. The noninvasive character of this examination with its high reproducibility makes it excellent for monitoring patients, and it helps the urologist decide whether to intervene surgically or continue a watchful waiting approach.¹³

Diuretic renography was introduced into clinical practice in 1978¹³⁵ and may be used to distinguish between upper tract dilatation with and without underlying obstruction. Following administration of radioisotope, when the isotope appears in the renal pelvis, a loop diuretic such as furosemide is given intravenously. If stasis is causing dilatation, the induced diuresis may result in prompt washout of the tracer from the renal pelvis. By contrast, when dilatation is caused by obstruction, the washout does not occur.¹³⁶ Data should be interpreted visually and by quantitative measurement including the half-life ($T_{1/2}$) for the excretion of the tracer from the collecting system.¹³⁷ It is generally accepted that clearance of the isotope from the collecting system with $T_{1/2}$ less than 15 minutes is normal, and a $T_{1/2}$ of more than 20 minutes may indicate obstruction in adults. Renal excretion of the tracer with a $T_{1/2}$ between 15 and

20 minutes is considered equivocal. An absent or blunted diuretic response resulting from decreased kidney function or grossly dilated pelvis makes interpretation of the test difficult, limits its usefulness, and may require support tools to increase the diagnostic performance.¹³⁸ As diuretic renography was initially developed, applied, and validated in adults,¹³⁵ its use and interpretation in pediatric patients, especially infants, can be difficult as the classical variables of the diuretic renogram may not allow an estimate of the best drainage, as seen when the bladder is full and because the effect of gravity on drainage is incomplete in recumbent babies. Moreover, kidney function and pelvic volume can influence the quality of drainage, with misinterpretation easily taking place if the renal pelvis is large and does not allow proper drainage within the study time.¹³⁹ Differential kidney function can also be obtained from the initial phase part of this modality, which remains a robust measure, provided there is adequate background subtraction. Pitfalls pertain to different aspects of the study such as the drawing of regions of interest, particularly in infants, and the estimation of the interval during which differential kidney function is calculated. There is no definition of a “significant” reduction in differential kidney function, but it has arbitrarily been set to values of below 40% to 45% in most studies.¹⁵

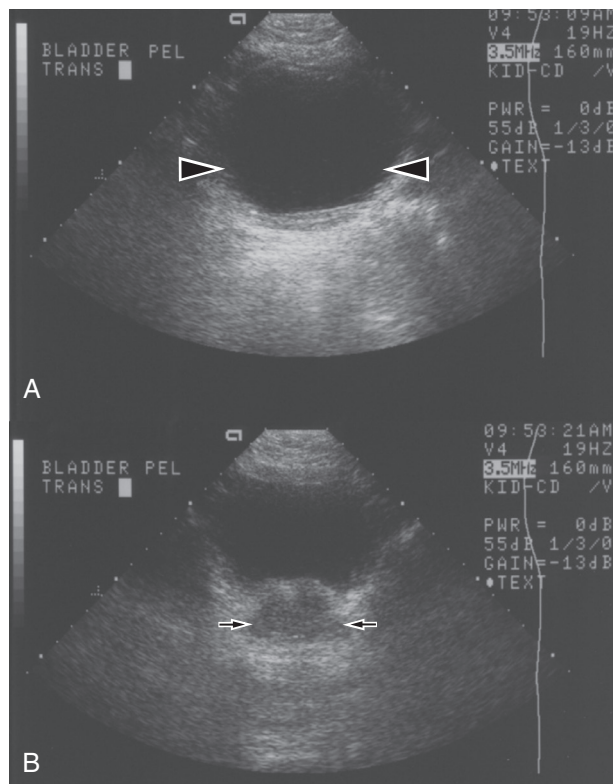


Fig. 39.5 Pelvic ultrasound. (A) Distended bladder (arrowheads). (B) Enlarged prostate (arrows), causing infravesical urinary obstruction.

POSITRON EMISSION TOMOGRAPHY

Experimentally, positron emission tomography permits quantitative imaging of the kidney at a spatial resolution appropriate for the organ. $H_2^{15}O$, $^{82}RbCl$, and $[^{64}Cu]$ ethylglyoxal bis(thiosemicarbazone) (ETS) are the most important radiopharmaceuticals for measuring renal blood flow.¹⁴⁰ Membrane organic cation and anion transporters are important for the function of the tubular epithelium and novel radiopharmaceuticals, such as copper-64-labeled mono oxo-tetra azamacrocyclic ligands and carbon-11-labeled metformin, have been used for molecular renal imaging^{140,141} and potentially may play a role in detection of kidney injury in response to urinary tract obstruction.¹⁴²

MAGNETIC RESONANCE IMAGING

MRI generates images by transmitting and receiving radiowave frequencies while altering the magnetic field strength to determine the spatial orientation and contrast of images. The energy used is nonionizing and therefore, as discussed earlier, the modality can safely be used in children and during pregnancy. However, MRI studies are relatively expensive, and acquisition times are relatively long compared with other modalities. A full MRI protocol can take up to 15 to 30 minutes, with any motion from the patient potentially degrading image quality. Sedation may therefore be necessary for children.¹⁴³

MRI sequences fall into T1-weighted or T2-weighted categories, with T2-weighted sequences being sensitive for fluids, which appear bright on the image, and hence protocols with multiple specialized T2-weighted sequences

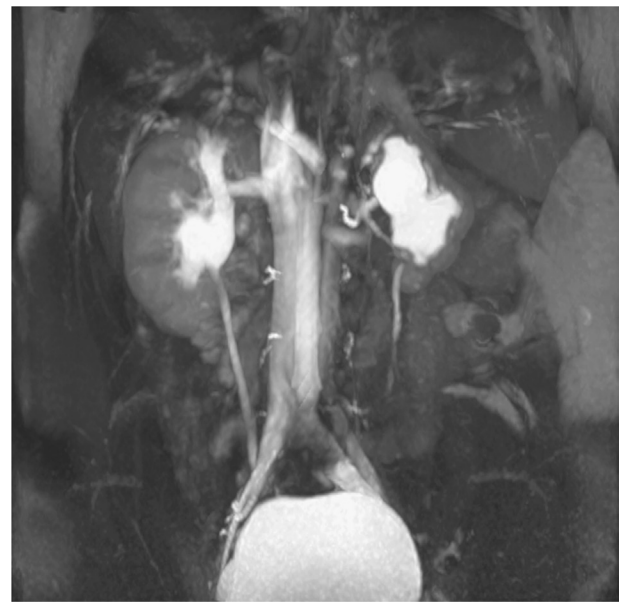


Fig. 39.6 Magnetic resonance (MR) imaging urography of left-sided hydronephrosis with parenchymal thinning. MR urographic image shows large dilatation of the left pelvicalyceal system and narrowing of the left ureteropelvic junction segment.

can be constructed to simulate a CT urogram (Fig. 39.6). As in CT, MRI sequences can be enhanced by administering an intravenous contrast agent.¹⁴³ Enhanced functional MRI studies can combine anatomic visualization of the kidneys and urinary tract with functional information acquired from the transit of the contrast from the cortex and medullary pyramids into the collecting system. This enables measurement of glomerular filtration rate; assessment of potential prognostic factors (hypoxia, inflammation, cell viability, degree of tubular function, and interstitial fibrosis); and monitoring of new therapies.¹⁴⁴⁻¹⁴⁶ Functional MRI can also provide valuable information regarding tissue energy consumption from so-called blood oxygenation level dependent (BOLD) imaging; these data may be helpful in the future in predicting the recovery of kidney function following obstruction.^{147,148}

Studies in children and adults have shown high sensitivity and specificity for magnetic resonance urography in diagnosing obstruction; however, identifying cause of obstruction was only possible in approximately 50%.¹⁴⁹ This is especially true for stone disease because MRI cannot directly detect calcifications or stone material.¹⁵⁰

Concerns related to the use of MRI in patients with kidney disease were highlighted by the risk of developing nephrogenic systemic fibrosis (NSF) induced by the toxicity of several gadolinium-based contrast agents in patients with severely impaired kidney function, although newer formulations of gadolinium do not appear to contribute to NSF.¹⁵¹

WHITAKER TEST

The Whitaker test is essentially a urodynamic evaluation of the upper urinary tract in which the hydrostatic pressure of the renal pelvis is measured while infusing a mixture of saline and contrast medium at a constant through a two-way nephrostomy tube, connected to a pressure transducer.¹⁵²

The patient is placed in the prone position on a fluoroscopic table, with a bladder catheter in situ, to ensure an empty bladder and vesical pressure measurement. The infusion starts at 10 mL/min and may be increased incrementally. The urinary tract is considered nonobstructed if renal pelvic pressure is less than 15 cm H₂O, equivocal at a pressure between 15 and 22 cm H₂O, and obstructed if pressure exceeds 22 cm H₂O.¹⁵² Due to its invasive nature and the availability of other noninvasive modalities, the Whitaker test is rarely used in modern clinical practice and is presented here largely for historical interest.¹⁵³

RETROGRADE AND ANTEGRADE PYELOGRAPHY

When other tests do not provide adequate anatomic detail or when obstruction must be relieved (e.g., obstruction of a solitary kidney, bilateral obstruction, and symptomatic infection in the obstructed system), more invasive investigation, with a combination of treatments, may be necessary. When retrograde pyelography is performed, this takes place during cystoscopy, by cannulating the ureteral orifice and injecting contrast.¹⁵⁴⁻¹⁵⁶ In some cases of complete obstruction, contrast may not reach the kidney, but the procedure will define the lower level of the obstruction. Retrograde pyelography can be combined with placement of a ureteral stent to relieve an obstruction or with possible stone extraction. Because the catheter passes through the bladder to reach the upper urinary tract, the risk of introducing infection must be kept in mind. Antegrade pyelography is performed by percutaneous cannulation of the renal pelvis and injection of the contrast material into the kidney and ureter.^{155,156} This procedure should establish the proximal level of obstruction and may also serve as a first step in relieving obstruction by means of percutaneous nephrostomy (Fig. 39.7).

PATHOPHYSIOLOGY OF OBSTRUCTIVE NEPHROPATHY

Even though acquired obstructive nephropathy in humans usually results from partial urinary tract obstruction and is generally prolonged in its time course, most mechanistic studies of kidney function in acquired obstruction use models of acute complete obstruction, usually for 24 hours. In these animal models, the extent of obstruction is clear and reproducible and, if the kidneys are studied soon after the obstruction is performed or released, the results are not confounded by changes in kidney structure brought on by inflammation or fibrosis. Complete obstruction of short duration strikingly alters renal blood flow, glomerular filtration, and tubular function while producing minimal anatomic changes in blood vessels, glomeruli, and tubules. However, in a more chronic setting, complete obstruction is characterized by a complex array of pathophysiologic processes leading to end-stage kidney disease (Fig. 39.8).¹⁵⁷

EFFECTS OF OBSTRUCTION ON RENAL BLOOD FLOW AND GLOMERULAR FILTRATION

Obstruction profoundly alters all components of glomerular function. The disruption in GFR depends on factors such as obstruction severity, duration, unilateral or bilateral nature,

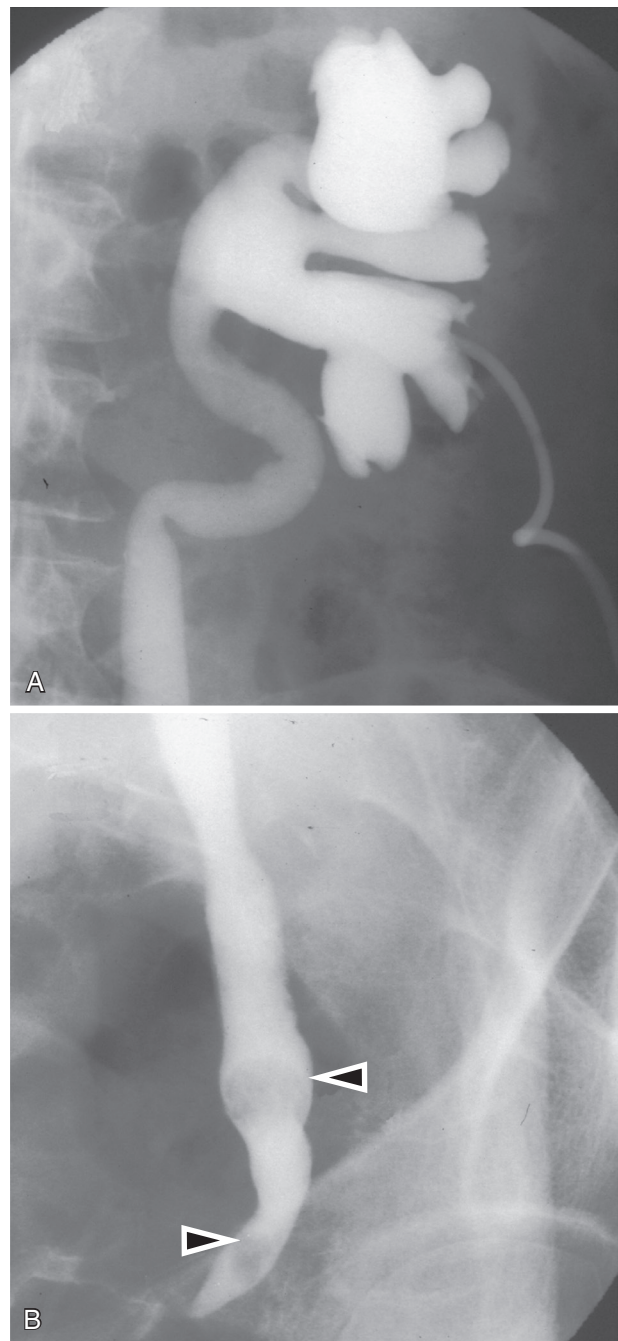


Fig. 39.7 Antegrade pyelography. (A) Dilated renal pelvis and calices on left. (B) Stones (arrowheads) as filling defects in the distal ureter (not seen on plain film). Intravenous pyelography was unsuccessful owing to the obstructed and malfunctioning kidney.

and whether the obstruction persists or is relieved.² To describe the effects of obstruction on glomerular filtration, we must review aspects of normal GFR. Whole-kidney GFR depends on functional glomerular filtration across all glomeruli and the proportion of glomeruli actually filtering. As detailed in Chapter 3, single-nephron GFR is determined by the glomerular blood flow, net ultrafiltration pressure across the glomerular capillary, and ultrafiltration coefficient (K_f). Glomerular blood flow and the hydraulic pressure in the glomerular capillary depend on afferent and efferent arteriolar resistances. Net ultrafiltration

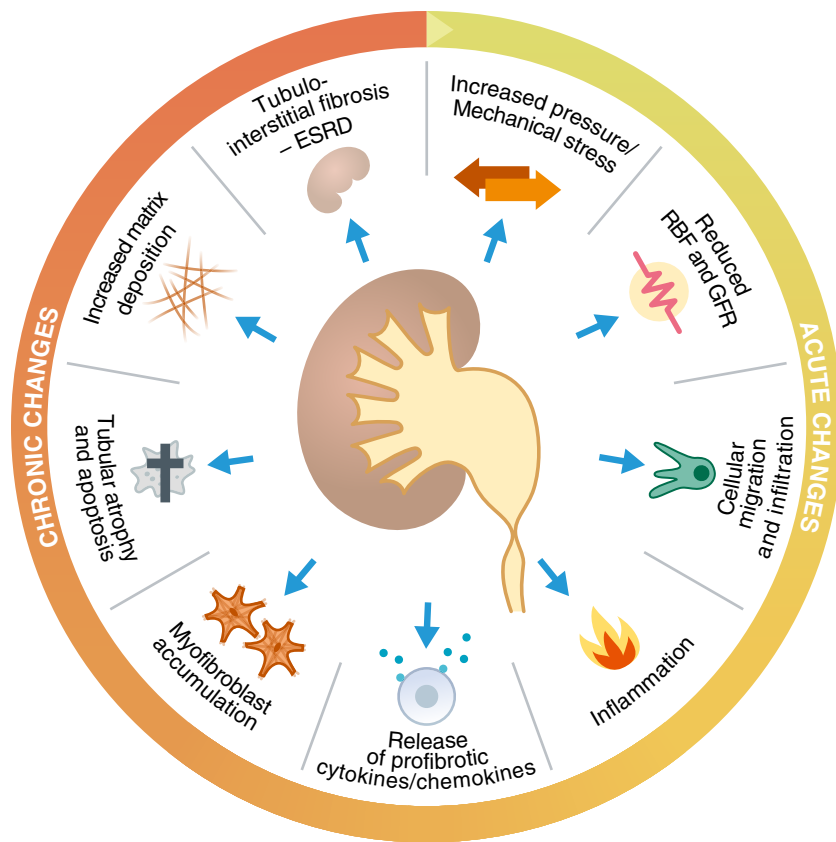


Fig. 39.8 Hallmarks of obstructive nephropathy showing key characteristics of functional and cellular consequences of urinary tract obstruction as a time lapse from acute to more chronic conditions. ESRD, End-stage renal disease; GFR, glomerular filtration rate; RBF, renal blood flow. (Figure modified related to Nørregaard R, Mutsaers HAM, Frøkær J, Kwon TH. Obstructive nephropathy and molecular pathophysiology of renal interstitial fibrosis. *Physiol Rev.* 2023;103:2827–2872.)

pressure is determined by glomerular capillary pressure, the hydraulic pressure of the Bowman space (proximal tubule hydraulic pressure), and the differences in oncotic pressure between the glomerular capillary and Bowman space. Kf is determined by the permeability properties of the filtering surface and the surface area available for filtration. Obstruction can modify any or all determinants, reducing GFR.

THE EARLY, HYPEREMIC PHASE

In the immediate 2 to 3 hours after the onset of ureteral obstruction, blockade of antegrade urine flow markedly increased proximal tubule hydraulic pressure. This increase in pressure in the Bowman space would be expected to halt GFR immediately.¹⁵⁸⁻¹⁶⁰ However, during this early phase of obstruction, the afferent arterioles dilate, decreasing afferent arteriolar resistance, increasing glomerular capillary pressure, and counteracting the increase in proximal tubule hydraulic pressure.^{158,159} Because this vasodilator or “hyperemic response” occurs in denervated kidneys in situ and in isolated perfused kidneys,^{161,162} it must result from intrarenal mechanisms. In fact, glomeruli of individual nephrons exhibit the same response in in vivo micropuncture experiments when antegrade urine flow is blocked by placement of a wax block in the tubule of the nephron.¹⁶³

Many mechanisms may mediate this afferent vasodilatation, including increases in vasodilatory hormones such as prostaglandins, reduced NaCl concentration at the macula densa, and a direct myogenic reflex.¹⁶⁴ This hyperemic response is not attenuated by renal nerve stimulation or

infusion of catecholamines¹⁶⁵ and may be linked to changes in interstitial pressure.¹⁶²

In the tubuloglomerular feedback response, reduced tubular flow past the macula densa induces reductions in afferent arteriolar resistance and increases in glomerular capillary pressure, so single-nephron GFR rises. Similarly, because obstruction reduces urine flow past the macula densa, it would be predicted to induce afferent vasodilatation.¹⁶³ However, elegant micropuncture studies separated the stoppage in flow from increases in the proximal tubule by placing an additional puncture in the tubule that was proximal to the blockage of flow to the macula densa. In this setting, flow past the macula densa was halted but proximal tubule remained normal because accumulating tubular fluid was permitted to leak out.¹⁵⁹ In such nephrons the increase in glomerular capillary pressure observed in obstructed tubules did not occur, indicating that the obstruction itself and not the macula densa stimulates afferent vasodilatation.¹⁵⁹

Renal prostaglandins and renal nerves play important roles in the hyperemic response. Indomethacin blocks the hyperemic response, indicating that vasodilator prostaglandins are critical to afferent vasodilatation.^{160,166} A renorenal reflex mechanism in the hemodynamic response to obstruction can be discerned from studies in bilateral obstruction, where the afferent vasodilatation response is absent or markedly attenuated.^{2,162} Obstruction of the left kidney augments afferent renal nerve activity from the left kidney and efferent nerve activity to the right kidney. Increased efferent nerve activity to the right kidney was accompanied by reduced blood flow to that kidney. This vasoconstrictor response was ablated by denervation of

either the left or right kidney before induction of left ureteral obstruction, suggesting that increased afferent renal nerve traffic triggers vasoconstrictive renorenal reflex activity that counteracts the early intrinsic renal vasodilator effects of obstruction in bilateral ureteral obstruction.¹⁶²

THE LATE, VASOCONSTRICTIVE PHASE

Because obstruction results in cessation of glomerular filtration, efforts to study the regulation of single-nephron GFR later in obstruction have measured determinants of GFR immediately after release of obstruction.^{2,167} Using this approach, investigators have shown that renal blood flow declines progressively after 3 hours of unilateral obstruction and through 12 to 24 hours of obstruction.^{168,169} Interestingly, although tubular pressures rise initially after obstruction, they then decline, so by 24 hours renal plasma flow, GFR, and intratubular pressures have all dropped below normal values.^{158,160,169,170} At 24 hours into the obstruction, examination of regional blood flow in the kidney by injections of silicone rubber reveals large areas of the cortical vascular bed that are either underperfused or not perfused at all.^{2,160,169} Depending on the species, the different vascular beds in the outer and juxtamedullary cortex receive differing proportions of the renal blood flow under basal conditions and following obstruction. However, at 24 hours of obstruction, reduced whole-kidney GFR is clearly due, in large part, to nonperfusion of many glomeruli.

Beyond 24 hours of obstruction, the single-nephron GFR of glomeruli that remain perfused is markedly decreased because of reduced blood flow to the afferent arteriole and afferent vasoconstriction, which, in turn, reduces glomerular capillary pressure.^{170,171} Because glomerular capillary pressure responds in the same manner when the individual nephron is blocked with oil for 24 hours before micropuncture measurements are performed, it is clear that afferent arteriolar vasoconstriction plays an important role in attenuating single-nephron GFR during the established phase of obstruction.¹⁷² These results indicate that, like the early hyperemic response, intrarenal mechanisms play the major role in the late vasoconstrictive response to ureteral obstruction. In bilateral obstruction, renal blood flow is reduced to levels 30% to 60% below normal (Table 39.1).^{170,171,173} In both unilateral and bilateral obstruction, single-nephron GFR falls to a similar degree. However, the mechanisms involved are different in the two conditions. In unilateral obstruction, reduced glomerular capillary pressure lowers the driving pressure for filtration when set against a nearly normal proximal tubular hydraulic pressure. By contrast, in bilateral obstruction, glomerular capillary pressure remains normal and GFR is halted by a highly elevated proximal tubular hydraulic pressure.¹⁷⁰ These results suggest that systemic factors, such as accumulation of extracellular fluid volume and urea, an increase in natriuretic substances, and alterations in renal nerve activity, modulate the vasoconstrictive effect of obstruction on the affected kidney.¹⁷³

REGULATION OF THE GLOMERULAR FILTRATION RATE IN RESPONSE TO OBSTRUCTION

The reduction in renal blood flow and GFR after release of obstruction varies with the species studied and

Table 39.1 Glomerular Hemodynamics in Ureteral Obstruction^a

Stage of obstruction	P_T	R_A	P_{GC}	SNGFR
1-2 hours unilateral	↑↑	↓	↑	=
24 hours unilateral	=	↑↑	↓	↓↓
24 hours bilateral	↑↑	=	=	↓↓
After release: 24 hours unilateral	↓	↑↑	↓↓	↓↓
After release: 24 hours bilateral	=	↑↑	↓	↓↓

^aSee text for discussion and references.

P_{GC} , Hydraulic pressure of Bowman space; P_T , proximal tubule hydraulic pressure; R_A , afferent arteriole resistance; SNGFR, single-nephron glomerular filtration rate; =, unchanged; ↑, increased; ↑↑, markedly increased; ↓, reduced; ↓↓, markedly reduced.

duration of obstruction.² Following release of a 24-hour complete unilateral obstruction, GFR remains below 50% of normal in dogs and 25% of normal in rats; renal blood flow remains markedly reduced in both species.² After release of bilateral ureteral obstruction, renal blood flow increases to levels higher than those observed after unilateral obstruction. This increase might be due to systemic natriuretic influences such as volume accumulation, reduced sympathetic tone, or increased circulating atrial natriuretic peptide. However, despite an increase in renal blood flow, GFR remains significantly impaired. The persistent reduction in GFR is due in part to nonperfusion or underperfusion of many glomeruli as shown in silicone rubber injection experiments.^{160,169} Where glomeruli remain perfused, strong afferent vasoconstriction reduces glomerular capillary pressure. This, combined with the decline in proximal tubule hydraulic pressure after release of obstruction, leads to a low driving force for glomerular filtration.^{170,171} In addition, a sharp reduction in K_f further contributes to the decline in GFR upon release of both unilateral and bilateral obstruction.^{170,171}

Several mechanisms contribute to afferent vasoconstriction and a reduced K_f. First, release of obstruction strikingly augments the flow of tubular fluid past the macula densa. Although the absolute rate of flow is still far below normal, the macula densa likely senses the change in flow, which may lead to intense vasoconstriction.² In favor of this view, the sensitivity of the tubuloglomerular feedback mechanism is enhanced in unilateral, as compared with bilateral, obstruction, suggesting that the ability of the mechanism to regulate afferent arteriolar tone is modulated by the extrarenal hormonal milieu.¹⁷⁴

There is substantial evidence that increased intrarenal secretion of angiotensin II participates actively in afferent vasoconstriction and reduced K_f following release of ureteral obstruction. Ureteral obstruction rapidly increases renal vein renin levels at a time when renal blood flow is normal or elevated, but at later time points, renal vein renin levels return to normal.¹⁷⁵⁻¹⁷⁷ In addition,

infusion of captopril, an angiotensin-converting enzyme (ACE) inhibitor, attenuated the declines in renal blood flow and GFR observed in both unilateral and bilateral obstruction.^{175,177} The effect of captopril persisted regardless of whether kinin activity was eliminated through infusions of either carboxypeptidase B, which destroys kinins, or aprotinin, which blocks kinin generation. This suggests that captopril reduced afferent arteriolar resistance primarily by blocking generation of angiotensin II rather than affecting kinins.² The significance of the renin-angiotensin system as an important contributor to vasoconstriction has recently been highlighted in studies where AT1 receptor blocker treatment attenuated the reduction in GFR in the postobstructive period in both adult rats¹⁷⁸ and rats with neonatally induced unilateral partial obstruction in response to long-term AT1 receptor antagonist treatment.¹⁷⁹

Thromboxane A2 (TXA2) plays a role in obstruction-induced vasoconstriction.^{175,180} Chronically, hydronephrotic kidneys exhibit increased TXA2 accumulation, as measured by accumulation of its more stable metabolite TXB2.¹⁸⁰ Furthermore, whole-kidney GFR and renal blood flow were increased in response to thromboxane synthase inhibitor treatment,^{175,181} likely by reducing afferent arteriolar resistance and thereby increasing Kf.¹⁸² From these results, TXA2 appears to be generated in the kidney after release of obstruction and mediates afferent vasoconstriction and reductions in Kf.

Although the source of TXA2 generation remains unclear in some cases,¹⁸³ but not all,¹⁸⁴ glomeruli isolated from obstructed kidneys have shown increased ability to synthesize TXA2 and other studies have suggested inflammatory cells as the source of TXA2. This is consistent with the observations that suppressor T cells and macrophages migrate to the renal cortex and medulla during the first 24 hours of obstruction, reaching levels fifteen fold higher than those observed in normal kidneys¹⁸⁵ and a parallel rise in TXA2 release and the fall in GFR.¹⁸⁵ These changes can be attenuated by renal irradiation, suggesting that obstruction stimulates migration of inflammatory leukocytes, which, in turn, generate vasoconstrictors such as TXA2.^{185,186} The role of angiotensin II in this process is highlighted because glomeruli isolated from obstructed kidneys showed increased eicosanoid synthesis after angiotensin II stimulation. In addition, treatment of animals with urinary tract obstruction with ACE inhibitors enhanced GFR and reduced TXA2 generation by glomeruli isolated from these animals.¹⁸⁷ Thus these vasoconstrictors may contribute to regulate afferent arteriolar resistance and GFR after release of obstruction.

Less severe vasoconstriction in animals with bilateral ureteral obstruction, as noted earlier, suggests that extrarenal factors play a major role in modulating the hemodynamic response of the kidney to obstruction and release of obstruction. In addition to renorenal reflexes, factors including volume accumulation, urea, atrial natriuretic peptide, and other natriuretic substances appear to mitigate vasoconstriction in response to bilateral obstruction.^{188,189} After 24 hours of obstruction, GFR is preserved to some degree if the contralateral kidney is also obstructed or removed.¹⁸⁸ In

addition, reinfusion urea, salt, and water content of the urine from the contralateral kidney markedly increase GFR in animals after release of 24 hours of unilateral obstruction.^{188,190} Taken together, this suggests that atrial natriuretic peptide, urea, and other excreted urine solutes have a protective effect and can ameliorate vasoconstriction post release of ureteral obstruction by direct vasodilatation of afferent arterioles, constriction of efferent arterioles, and an increase in Kf.

Additional studies in dogs and rats have implicated endothelins as contributors to reduced GFR in obstruction, while prostaglandin E2 (PGE2) and nitric oxide (NO) help alleviate glomerular vasoconstriction in the chronic obstructed kidneys.^{191,192} Renal PGE2 levels increase markedly in obstruction (see later) and in states of extracellular volume expansion, as occurs in bilateral ureteral obstruction. Given the vasodilator effects of PGE2, it appears likely that increased levels could ameliorate falls in GFR in obstruction. Bilateral obstruction may reduce generation of NO, leading to a net vasoconstrictive effect.¹⁸⁷

PUUO is associated with increased renin-angiotensin-aldosterone system activity, elevated oxidative stress, reduced NO bioavailability, and sensitized afferent arteriolar reactivity and renal autoregulation, leading to blood pressure elevation in both mice and rats.¹⁹³ Renal denervation in PUUO rats attenuated both hypertension and salt sensitivity and normalized the renal excretion pattern, whereas the degree of renal fibrosis and inflammation was not changed. This suggests a link among renal nerves, increased blood pressure, and modulation of nicotinamide adenine dinucleotide phosphate oxidase function.

In summary, both intrarenal and extrarenal factors combine to profoundly decrease GFR during and immediately after release of obstruction. The decrease in GFR is caused by a sharp reduction in the number of perfused glomeruli and by a reduction in the single-nephron GFR of functioning nephrons. Decreased Kf and increased afferent arteriolar resistance reduce single-nephron GFR. Increases in various vasoconstrictors, such as angiotensin II and TXA2, as well as other vasoconstrictors, some coming from inflammatory cells, augment these hemodynamic effects. In the setting of bilateral obstruction, retention of urea and other solutes, as well as volume expansion and increases in circulating levels of vasodilators such as atrial natriuretic peptide, helps offset these vasoconstrictive effects, but only partially.

RECOVERY OF GLOMERULAR FUNCTION AFTER RELIEF OF OBSTRUCTION

The extent of recovery of glomerular filtration after release of obstruction depends on several factors including the duration and extent of obstruction, presence or absence of a functioning contralateral kidney, presence or absence of associated infection, and level of preobstruction renal blood flow.^{2,194} In a classic experiment in dogs subjected to a 1-week period of complete UUO, GFR fell to 25% of normal on release of the obstruction and recovered gradually to 50% of normal levels 2 years later, indicating persisting irreversible changes.¹⁹⁵ In rats, release of UUO

of 7- and 14-day duration left residual GFR at 17% and 9% of control levels, respectively, when the contralateral kidney was left in place, and at 31% and 14% when the animals underwent contralateral nephrectomy at the time of release of the obstruction.¹⁹⁶ A similar beneficial effect on the obstructed kidney of contralateral nephrectomy was observed in rats subjected to chronic partial obstruction.¹⁹⁶ As discussed earlier, this beneficial effect likely results from the accumulation of urea and other solutes and increased levels of ANP when the functioning contralateral kidney is absent.

The partial recovery of total GFR after release of obstruction masks an uneven distribution of blood flow and nephron function. In micropuncture studies, some nephrons never regain filtration function, whereas others reveal striking hyperfiltration.¹⁹⁴ It appeared in some studies that surface nephrons exhibited normal single-nephron GFR, whereas the whole-kidney GFR was reduced to 18% of normal.¹⁹⁷ These results suggest that chronic partial obstruction causes selective damage to juxtamedullary and deep cortical nephrons.^{169,194,197} Similarly, studies of the long-term outcome of complete 24-hour ureteral obstruction revealed that total GFR recovered to normal levels by 14 and 60 days after release of obstruction. However, 15% of the glomeruli were not filtering in recovered kidneys, and other nephrons were hyperfiltering. In this model of complete obstruction, there appeared to be no selective advantage for surface glomeruli over deep cortical and juxtamedullary glomeruli.¹⁹⁴

Similarly, in the developing kidney, the duration of obstruction and timing of release significantly impact long-term renal function. Releasing obstruction after

1 week prevented hydronephrosis and reduced renal blood flow and GFR impairment in rats with PUUO from birth. In contrast, releasing obstruction after 4 weeks resulted in minimal or no kidney function in the neonatal obstructed kidney, highlighting the importance of early release of obstruction for optimal protection of kidney function after the maturation process is completed.¹⁹⁸ Similarly, neonatal unilateral partial obstruction in pigs demonstrated that impaired nephrogenesis resulted in a reduced number of glomeruli in the obstructed kidney.¹⁹⁹ Preserved whole-kidney function suggests some degree of glomerular hyperfiltration. In line with the hypothesis that hyperfiltration is associated with an increased risk of systemic hypertension,²⁰⁰ studies in pigs and rats have demonstrated that renal expressions of neuronal nitric oxide synthase (NOS) and endothelial NOS proteins were lower in animals with hydronephrosis.^{201,202} These findings suggest that the reduced NO response in the obstructed hydronephrotic kidney and subsequent resetting of the tubuloglomerular feedback mechanism play an important role in the development of hypertension in hydronephrosis.

EFFECTS OF OBSTRUCTION ON TUBULE FUNCTION

Obstruction severely impairs the ability of renal tubules to transport Na^+ , K^+ , and H^+ and reduces their ability to concentrate and dilute the urine (Table 39.2).^{2,203-209} The resulting inability to reabsorb water and solutes facilitates postobstructive diuresis and natriuresis. As in the case with glomerular filtration, the extent of disruption of tubular transport depends directly on the duration and severity of the obstruction. Pathologically, prolonged obstruction leads to profound tubular atrophy and chronic interstitial

Table 39.2 Segmental Reabsorption in Superficial and Juxtamedullary Nephrons and in Collecting Ducts in Normal Rats After Release of Bilateral or Unilateral Obstruction

Site	Normal		After unilateral obstruction		After bilateral obstruction	
	Water remaining (%)	Na^+ remaining (%)	Water remaining (%)	Na^+ remaining (%)	Water remaining (%)	Na^+ remaining (%)
S_1	100	100	100	100	100	100
S_2	44	44	26	26	45	45
S_3	26	14	21	12	40	22
S_4	9.4	5	3.2	1.9	25	7
J_1	100	100	100	100	100	100
J_2	12	40	42	52	42	62
CD_1	3.3	2	4.2	3.8	8	6
CD_2	0.4	0.6	2.9	2.5	16.7	12

In obstruction, increased proportions of filtered salt and water are delivered to the loop of Henle in juxtamedullary nephrons (J_1 and J_2 indicating decreased reabsorption). Delivery of sodium and water to the first accessible portion of the inner medullary collecting duct, labeled CD_1 , was also increased, and net sodium and water reabsorption along the inner medullary collecting duct (between CD_1 and CD_2) was diminished in both bilateral and unilateral obstruction. In bilateral obstruction, there was net addition or secretion of sodium and water into the lumen of the inner medullary collecting duct, suggesting that in this setting the inner medullary collecting duct secretes sodium and water.

CD_1 , Collecting duct at base of papilla, first accessible portion of inner medullary collecting duct; CD_2 , end of collecting duct as it opens into renal pelvis; J_{1-2} , values found in juxtamedullary nephrons; J_1 , Bowman space; J_2 , tip of loop of Henle; S_{1-4} , values found in superficial nephrons; S_1 , Bowman space; S_2 , end of the proximal convoluted tubule; S_3 , earliest portion of the distal tubule; S_4 , end of the distal tubule/beginning of collecting duct.

(Data from Harris KP, Schreiner GF, Klahr S. Effect of leukocyte depletion on the function of the postobstructed kidney in the rat. *Kidney Int.* 1989;36:210–215.)

Table 39.3 Function of Isolated Perfused Tubules in Obstructive Nephropathy

	J_v SPCT (nL/mm/min)	J_v PST (nL/mm/min)	ΔCl^- MTAL (mEq/L)	J_v CCT (ADH) (nL/mm/min)
Control	0.75 ± 0.08	0.25 ± 0.02	-37 ± 3	0.90 ± 0.08
Unilateral obstruction	0.73 ± 0.11	0.12 ± 0.03	-9 ± 1	0.22 ± 0.04
Bilateral obstruction	0.80 ± 0.08	0.16 ± 0.02	-10 ± 1	0.23 ± 0.04

The J_v in the SPCT was not affected by obstruction, whereas J_v in PST decreased by 52% in unilateral obstruction (0.12 ± 0.03 vs. 0.25 ± 0.02 nL/mm/min) and similarly in response to bilateral obstruction. In MTAL the ability to lower the perfusate chloride ion concentration was reduced by 76% (-9 ± 1 vs. -37 ± 3 mEq/L) and similarly in response to bilateral ureteral obstruction. After relief of unilateral obstruction, the ability of the CCT to respond to ADH was reduced by 76% (0.22 ± 0.04 vs. 0.90 ± 0.08 nL/mm/min) and similarly following relief of bilateral obstruction.

ADH, Antidiuretic hormone; CCT, cortical collecting tubule; J_v , net fluid reabsorption rate per length of the tubule segment; PST, cortical proximal straight tubule; SPCT, superficial proximal convoluted tubule; ΔCl^- MTAL, change in Cl^- concentration per length of the medullary thick ascending limb.

(Data from Reyes AA, Martin D, Settle S, Klahr S. EDRF role in renal function and blood pressure of normal rats and rats with obstructive uropathy. *Kidney Int.* 1992;41:403-413.)

inflammation and fibrosis (see the following section), whereas at early time points after the onset of obstruction, such as at 24 hours, there are only slight structural and ultrastructural changes including mitochondrial swelling, modest blunting of basolateral interdigitations in the thick ascending limb and proximal tubule epithelial cells, flattening of the epithelium, and some widening of the intercellular spaces in the collecting ducts.^{2,210,211} The only cell death at early time points is observed at the tip of the papilla, where focal necrosis may be observed.²¹⁰ Because there is so little cell damage and the model is simple, most investigators have examined the effect of 24 hours of complete ureteral obstruction on tubular function. As discussed later, regulation of tubular transport is complex and due to both direct damage of epithelial cells and the action of extra tubular mediators, arising from the kidney and extrarenal sources.

EFFECTS OF OBSTRUCTION ON TUBULAR SODIUM REABSORPTION

Following release of 24 hours of UUO, volume excretion from the postobstructed kidney is normal or slightly increased^{2,173,188,212} (see Table 39.2). However, as discussed earlier, normal volume excretion occurs in the setting of a markedly reduced (20% of normal) GFR. This leads to a higher fractional excretion of sodium (FENa). After release of bilateral obstruction, salt and water excretion rises from fivefold to ninefold higher than normal.^{2,173,203,204} Given the decreased GFR in this setting, FENa can even be 20-fold higher than normal.

The micropuncture studies summarized in Table 39.2 demonstrate that the reabsorption defect after release of obstruction is localized similarly in both unilateral and bilateral ureteral obstruction. Obstruction reduced net salt and water reabsorption in the medullary thick ascending limb (MTAL), distal convoluted tubule, and entire length of the collecting duct including its cortical, outer medullary, and inner medullary segments.²⁰³

These studies in whole animals were confirmed and extended by a series of studies from multiple laboratories using isolated perfused tubule and cell suspension preparations (Table 39.3). As shown, the segments

including proximal straight tubule, MTAL, and cortical collecting duct isolated from unilaterally or bilaterally obstructed animals exhibited profound impairment of sodium reabsorptive capacity.^{204,205} This finding was confirmed in studies of freshly prepared suspensions of MTAL cells from obstructed kidneys, in which transport-dependent oxygen consumption, a measure of sodium reabsorptive capacity, was markedly reduced.²⁰⁶ Given the regulating role of mineralocorticoids in the collecting duct, reductions in collecting duct sodium reabsorption capacity are observed in tubules taken from obstructed kidneys, regardless of prior mineralocorticoid treatment.^{205,207,208} As the inner medullary collecting duct is challenging to study directly, investigations using cell suspensions show a marked reduction in transport-dependent oxygen consumption in inner medullary collecting duct cells isolated from animals with bilateral obstruction.²⁰⁹

Taken together, the data derived from micropuncture, tubule perfusion, and cell suspension studies reveal a striking impairment of sodium reabsorption in the proximal straight tubule, MTAL, and entire collecting duct. Because these functional derangements occur in the absence of clear-cut ultrastructural damage to the epithelial cells, obstruction likely induces a selective impairment in the regulation of active cellular transport mechanisms. Unlike the situation with glomerular filtration, the functional impairment of tubules appears similar in both unilateral and bilateral obstruction.^{205,208,209} Thus it appears that a major component of impaired active transport is likely due to direct tubular cell injury rather than continuous action of natriuretic substances. Added to this intrinsic injury, natriuretic substances may be responsible for the apparent secretion of sodium and water in the inner medullary collecting duct of animals after release of bilateral obstruction (see Table 39.2).

A combination of studies of cell suspensions and antibody-based targeted proteomics where long-term regulation of renal transporters and channels can be examined in intact animals to understand the integrated response to obstruction has improved the molecular understanding of mechanisms by which tubular epithelial cell sodium

reabsorption is impaired in the setting of obstruction. Active tubular Na^+ transport requires an apical entry step (e.g., NKCC2 cotransporter in MTAL or epithelial Na^+ channels [ENaCs] in the collecting duct) coupled to the basolateral Na^+ , K^+ -ATPase. In addition, the cell must generate sufficient adenosine triphosphate (ATP) to fuel active transport by the ATPase. Suspensions of MTAL cells from obstructed kidneys exhibited markedly reduced furosemide-sensitive oxygen consumption,²⁰⁶ indicating striking decreases in apical NKCC2 cotransporter activity in these cells. Isotopic bumetanide binding revealed a marked reduction in the number of cotransporter protein molecules available for binding on the membrane, with no change in affinity of binding, indicating that obstruction downregulates the expression of the cotransporter protein on the membrane surface.²⁰⁶ More recent studies using antibody-based targeted approaches clearly showed that obstruction diminishes expression of the cotransporter protein on the MTAL cell apical membrane.²¹³ Similar approaches demonstrated downregulation of Na^+ , K^+ -ATPase of both α - and β -subunits at the transcriptional and posttranscriptional levels.²¹³

In the inner medullary collecting duct, similar studies demonstrated downregulation of ENaC.²¹⁴ Consistent with this, suspensions from obstructed kidneys showed marked decreases in amiloride-sensitive oxygen consumption, as well as amiloride-sensitive isotopic sodium entry into hyperpolarized cells.²⁰⁹

Obstruction reduces oxygen consumption and ATPase activity in inner medullary collecting duct cells, similar to MTAL cells. In addition, the levels of both Na^+ , K^+ -ATPase subunits decrease in these cell preparations.²⁰⁹ Patterns of messenger RNA (mRNA) expression resemble MTAL, indicating that the pump subunits can be downregulated at both transcriptional and posttranscriptional levels. Using the targeted antibody-based approach, it has been demonstrated that in both unilateral and bilateral ureteral obstruction, the expressions of Na^+ / H^+ exchanger (NHE3) and the Na^+ / PO_4^{3-} exchanger (NaPi-2) were strikingly decreased in the proximal tubule.^{213,215} These changes occurred in both the proximal convoluted and proximal straight tubule, even though the micropuncture and tubule perfusion studies cited earlier demonstrated preserved proximal convoluted tubule salt reabsorption and inhibition of proximal straight tubule reabsorption.^{215,216} Transporter protein and apical membrane expression of the distal convoluted tubule Na^+ / Cl^- cotransporter are also reduced, implying similar obstruction effects as seen in the MTAL and collecting duct.^{213,215}

Collectively, these findings reveal that obstruction leads to decreased membrane expression of proteins involved in sodium transport. In addition, metabolic studies suggest a decrease in oxidative and glycolytic enzyme activities, implying reduced energy production in these cells. This might be intensified by decreased basolateral folding and mitochondrial density in obstructed kidney tubules.²¹⁰ Notably, in MTAL and collecting duct suspensions, obstruction reduces oxygen consumption linked to transport, implying ATP generation is not the limiting factor for active transport. Thus the decline in epithelial sodium transport due to obstruction seems to be a controlled response tied to lowered metabolic demands during obstruction.

The mechanisms and pathways responsible for down-regulation of transport proteins in tubular epithelial cells by obstruction remain largely incomplete. Possible signals include the halting of urine flow, increased hydrostatic pressure on tubular epithelial cells, changes in blood flow to the tubules or in interstitial pressure, and generation of natriuretic substances in the kidney that result in long-term inhibition of transporter function. Powerful mass spectrometry analysis of tissue from obstructed rat kidneys and mpkCCD cells has led to proteomic identification of significant changes in more than 100 proteins including those belonging to the cytoskeleton.²¹⁷ These findings suggest that obstruction induces acute molecular changes in the renal cytoskeleton, in part, mediated by increased stretch of the renal tubular cells during obstruction.²¹⁷

Obstruction disrupts glomerular filtration, causing a dramatic reduction (or even cessation) in urine production. This lowers sodium delivery to tubular segments, resulting in sluggish apical membrane Na^+ entry due to unfavorable electrochemical gradients. Reduced Na^+ entry might then directly stimulate downregulation of transporter activity and expression. In MTAL and inner medullary collecting duct cells, blocking Na^+ entry by furosemide or amiloride, respectively, promptly reduced ouabain-sensitive oxygen consumption,^{206,218} indicating acute downregulation of Na^+ , K^+ -ATPase. In addition, in mineralocorticoid-clamped animals, chronic Na^+ entry blockade at the MTAL or cortical collecting duct further reduced ouabain-sensitive ATPase in microdissected tubule segments.^{219,220}

This suggests that halted urine flow could be a primary signal for obstruction-induced Na^+ transport downregulation.²¹⁸ Testing this hypothesis, inhibiting apical Na^+ entry for 24 hours in A6 cells (mimicking cortical collecting duct cells) led to reduced apical sodium entry even after blockade removal.²²¹ This downregulation coincides with selective β -subunit decrease in ENaC expression on apical membranes of the A6 cells. In whole cells, α - and γ -subunits remain unaffected.²²² In rats with obstruction, α -, β -, and γ -subunits of ENaC are downregulated, suggesting all three subunits contribute to impaired sodium reabsorption.²¹⁴ In contrast to results in cell suspensions or whole kidney,^{206,209,213,215} inhibition of apical sodium entry had no effect on expression of either subunit of Na^+ , K^+ -ATPase.²²² This directly proves that reduced Na^+ entry rate (likely during blocked urine flow) directly downregulates renal epithelial cell Na^+ transport.

In addition to the direct effects of halting urine flow, changes in intrarenal mediators and subcellular pathways likely play a critical role in the reduction of salt transport observed with obstruction. Obstruction markedly accelerates the already rapid generation of PGE₂ in the renal medulla.^{180,181,184,223} The molecular basis for this is a dramatic medullary cyclooxygenase-2 (COX-2) induction.^{223,224} Consistent with the known effect of PGE₂ to markedly inhibit Na^+ reabsorption in the MTAL, as well as in the cortical and inner medullary collecting ducts,²²⁵⁻²²⁷ COX-2 inhibition in rats with obstruction and release of obstruction attenuated the downregulation of NHE2, NKCC2, and Na^+ , K^+ -ATPase.^{224,228} From these results, obstruction likely reduces apically localized sodium cotransport proteins in the tubule epithelium and sodium

pump activity in tubular epithelia in part by increasing renal levels of PGE₂.

As discussed earlier, obstruction leads to a monocellular infiltrate in the kidney,¹⁸⁵ mainly around the peritubular distribution.¹⁸⁵ When obstructed kidneys were exposed to radiation, medullary inflammation diminished and there was a modest reduction in fractional excretion of sodium.¹⁸⁶ In addition, obstruction increases renal angiotensin II generation, which might impact the regulation of renal sodium handling. Inhibiting the AT1 significantly lessened the downregulation of NHE3 and NKCC2, leading to a decrease in renal sodium loss.⁴⁸

In summary, obstruction reduces net reabsorption of sodium in several nephron segments including the proximal straight tubule, MTAL, and cortical and inner medullary collecting ducts by downregulating the expression and activities of specific transporter proteins. Several signals mediate this downregulation, such as the cessation of urine flow with its attendant reduction of the rate of Na⁺ entry across the apical membrane, increased levels of natriuretic substances such as PGE₂, and infiltration of the obstructed kidney by mononuclear cells.

When both ureters are obstructed, extrarenal factors markedly enhance the sodium-wasting tendency already present in the obstructed kidney. One mechanism involves the volume expansion that occurs when bilateral obstruction ablates all renal function. Volume expansion impairs activity in the sympathetic nervous system, reduces circulating levels of aldosterone along with reduced renal clearance, and increases levels of atrial natriuretic peptide. Reduced sympathetic tone and aldosterone, coupled with increased atrial natriuretic peptide, markedly stimulate sodium excretion. Atrial natriuretic peptide likely represents a particularly important mediator of salt wasting in bilateral obstruction. Levels of atrial natriuretic peptide are markedly elevated in bilateral but not unilateral obstruction.²²⁹ Atrial natriuretic peptide enhances salt wasting at several nephron segments. By blocking renin release in the macula densa and angiotensin action in the proximal tubule, atrial natriuretic peptide reduces proximal tubule sodium reabsorption.^{189,229,230} Atrial natriuretic peptide also reduces aldosterone release and directly inhibits sodium reabsorption in the collecting ducts.^{189,229,230} In agreement with this mechanism, infusion of atrial natriuretic peptide into animals in which obstruction has just been released leads to marked increases in sodium and water excretion.²²⁹ Moreover, efforts to reduce circulating atrial natriuretic peptide levels following bilateral obstruction attenuated sodium excretion somewhat.²²⁹

EFFECTS OF OBSTRUCTION ON URINARY CONCENTRATION AND DILUTION

Because obstruction reduces the ability of the renal tubules to concentrate and dilute the urine, urine osmolality following release of obstruction in humans and experimental animals approaches that of plasma.^{2,231,232} Dilution of the urine requires that the thick ascending limb reabsorbs sodium without water and that the collecting duct maintains the dilute urine by not reabsorbing water along its length, despite the presence of a concentrated medullary interstitium.²³³ Concentration of the urine requires active sodium reabsorption in the thick limb and the action of

the countercurrent multiplier to generate a concentrated medullary interstitium, as well as the ability of the collecting duct to insert the vasopressin-regulated water channel aquaporin-2 (AQP2) into the apical membrane.^{234,235}

Obstructive nephropathy disrupts several of these mechanisms.^{215,231,232,236} As noted earlier, obstruction also markedly reduces MTAL sodium reabsorption, limiting this segment's ability to dilute the urine and generate a high medullary interstitial osmolality. Indeed, interstitial osmolality has been shown to be reduced in obstructed kidneys.² In addition, collecting ducts isolated from obstructed kidneys reveal normal basal water permeabilities but a marked reduction in their ability to increase water permeability in response to antidiuretic hormone or other stimulants of cyclic adenosine monophosphate (cAMP) accumulation in the cells. As was the case with sodium transport, the effects were similar in unilateral and bilateral obstruction.^{231,236} Detailed mechanistic studies show that obstruction markedly reduces transcription of mRNA encoding AQP2, as well as synthesis of AQP2 protein, and that collecting duct cells in obstructed kidneys do not traffic AQP2-containing vesicles effectively to the apical surface in response to vasopressin or increased cAMP.^{231,235-237} Part of this failure in trafficking results from a decrease in phosphorylation of AQP2 in obstructed kidneys²¹⁵ and likely also due to the fact that V2 receptor protein expression is downregulated.²³⁸ Redistribution of AQP2 and AQP2 phosphorylated at ser261 to more intracellular localizations after bilateral obstruction and colocalization with the early endosomal marker EEA1 and the lysosomal marker cathepsin D suggest that early downregulation of AQP2 could in part be caused by degradation of AQP2 through a lysosomal degradation pathway.²³⁹ In addition, UUO markedly decreases synthesis and deployment to the basolateral membrane of aquaporin-3 and aquaporin-4; when AQP2 is in the apical membrane, these aquaporins mediate the water flux across the basolateral membrane.²¹⁵ Enhancing the causal relationship of the changes in aquaporin activity and ability to concentrate the urine, expression of AQP2 remains suppressed for 7 days following relief of the obstruction, and the rise in urinary concentration parallels the recovery in AQP2 expression.^{215,231,232,236}

The fact that collecting ducts from obstructed kidneys do not respond to cAMP indicates that the lesion also involves sites other than the receptor for antidiuretic hormone.²³⁸ Consistent with the hypothesis that PGE₂-mediated inhibition of collecting duct water permeability does not directly affect cAMP levels but may have post-cAMP effects rather than actions via cAMP regulation,²⁴⁰ previous studies have shown that cyclooxygenase-2 (COX-2) inhibition prevented dysregulation of AQP2 in obstructed kidneys where COX-2 protein expression was markedly increased (Fig. 39.8).^{224,241} Consistent with the COX-2 induction observed in the kidney inner medulla in bilateral ureteral obstruction (BUO), COX-2 mRNA and protein levels are also increased in response to UUO.^{242,243} Reactive oxygen species are shown to trigger this increase in rats with 3-day UUO, but antioxidants like nicotinamide adenine dinucleotide phosphate oxidase inhibitor and rotenone can counteract this effect.²⁴⁴ Treating with superoxide dismutase 2 (SOD2)-mimic manganese (III) tetrakis ([4-benzoic acid] porphyrin chloride [MnTBAP]) reduces AQP2 downregulation after 7

days of UO and also suppresses COX-2 induction.²⁴⁵ This finding suggests that mitochondrial oxidative stress partly mediates AQP2 downregulation in obstructive nephropathy through the COX-2 pathway.

On the basis of these results, it can be concluded that the defect in urinary dilution in obstruction is due to reduced ability of the thick ascending limb to dilute the urine by transporting salt from the lumen of the tubule to its basolateral side. The collecting duct in obstructed kidneys maintains its low water permeability in the absence of antidiuretic hormone, so the failure to dilute the urine is not due to collapse of osmotic gradients in the collecting duct. The inability to concentrate the urine results from the failure of the thick limb to generate a concentrated interstitium, as well as the inability of the collecting duct to synthesize and traffic AQP2 and other water channels in response to antidiuretic hormone.

EFFECTS OF RELIEF OF OBSTRUCTION ON URINARY ACIDIFICATION

Obstruction dramatically reduces urinary acidification capacity in experimental animals and humans. In humans, release of obstruction does not lead to bicarbonate wasting, indicating that proximal tubular bicarbonate reclamation is maintained. By contrast, in both experimental animals and patients following release of obstruction, the urine pH does not decrease in response to an acid load, indicating that obstruction impairs the ability of the distal nephron to acidify the urine.²⁴⁶⁻²⁴⁸ This defect likely involves proton transport proteins in the collecting duct^{246,247} and proximal tubule and thick ascending limb of Henle.^{248,249}

Reduced collecting duct acid secretion could result from defects in H^+ (H^+ -ATPase or H^+/K^+ -ATPase) or HCO_3^- (e.g., Cl^-/HCO_3^- exchange) transport pathways, back leak of protons down their electrochemical gradient from the lumen to the basolateral side of the tubule, or, in the cortical collecting duct, the failure to generate a sufficiently lumen-negative transepithelial voltage.^{246,247,250} As described in detail earlier, obstruction reduces the activity of apical ENaC in the cortical collecting duct; the resulting loss of luminal negativity may attenuate acid secretion in these segments.^{246,247}

In the rat inner medullary collecting duct (studied by micropuncture) and in isolated perfused rat and rabbit outer medullary collecting duct, obstruction markedly reduces luminal acidification rates.²⁴⁶ Because Na^+ transport does not play a major role in acidification in these segments, the defect must be due to direct inhibition of acid or HCO_3^- transport pathways, or back leak of protons from lumen to interstitium.²⁵⁰

Antibody-based targeted studies examining the Cl^-/HCO_3^- exchanger and subunits of the H^+ -ATPase revealed reduced expression of these transporters in collecting ducts of unilaterally obstructed compared with contralateral, and control kidneys.^{248,250} Two possible mechanisms of reduced acid secretion were explored.²⁵⁰ One was that the intercalated cells in obstructed kidneys would exhibit a high proportion of “reverse” orientation, with the proton pump in the basolateral membrane and the Cl^-/HCO_3^- exchanger in the apical membrane. The other possibility was that the orientation of intercalated cells would not change, but there would be reduced expression of the H^+

or HCO_3^- transporter. The orientation of the intercalated cells was not altered by obstruction. However, obstruction did reduce the appearance of H^+ -ATPase along the apical membranes of intercalated cells, without altering the total content of H^+ -ATPase in extracts of renal cortex or medulla, in unilaterally obstructed as compared with contralateral kidneys.²⁵⁰

In addition to defective collecting duct, H^+ transport-reduced generation of the main buffer that carries acid equivalents in the urine, ammonia has also been observed in kidneys released from obstruction. Cortical slices of obstructed kidneys exhibit reduced glutamine uptake and oxidation, reduced gluconeogenesis, and reduced total oxygen consumption, resulting in a reduced ability to generate ammonia from glutamine.^{251,252}

EFFECTS OF RELIEF OF OBSTRUCTION ON EXCRETION OF POTASSIUM

As with sodium excretion, potassium excretion increases markedly following release of bilateral obstruction.^{253,254} Micropuncture and microcatheterization studies show that proximal potassium reabsorption is unchanged by obstruction while potassium is more rapidly secreted in the collecting duct, likely due to increased distal delivery and therefore more rapid distal luminal flux of sodium and volume following release of obstruction.^{203,253} By contrast, after release of unilateral obstruction, potassium excretion falls roughly in proportion to the reduction in GFR,²⁵⁵ an effect that may be related to reduced distal delivery of sodium. However, administration of sodium sulfate in this state does not stimulate potassium excretion in obstructed kidneys as it does in controls, suggesting that collecting ducts in unilateral obstructed kidneys have an intrinsic defect in potassium secretion.²⁵⁶ This intrinsic defect may represent a response similar to the downregulation of sodium transporters in obstructed kidneys described in detail earlier. The kaliuretic effect observed in bilateral obstruction may well be due to the influence of elevated levels of atrial natriuretic peptide, which, at high levels, can stimulate potassium secretion in the distal nephron.

EFFECTS OF RELIEF OF OBSTRUCTION ON EXCRETION OF PHOSPHATE AND DIVALENT CATIONS

When bilateral ureteral obstruction is released, phosphate excretion rises in proportion to sodium excretion.^{213,215,257,258} Phosphate restriction before the release of the obstruction prevents phosphate accumulation during bilateral obstruction, thereby blocking the increase in phosphate excretion.²⁵⁷ This can also be achieved by blockade of angiotensin II-mediated effects, highlighting the importance of enhanced renal angiotensin II levels in the obstructed kidney.⁴⁸ By contrast, release of unilateral obstruction results in phosphate retention, likely due to reduced GFR and avid proximal phosphate reabsorption.²⁵⁹ Calcium excretion may be increased or decreased, depending on whether the obstruction is unilateral or bilateral and depending on the species studied.^{257,259} Magnesium excretion is markedly increased after release of either bilateral or unilateral obstruction. This magnesium wasting probably occurs because both forms of obstruction markedly attenuate thick ascending limb sodium reabsorption, leading to reduced positive luminal transepithelial voltages and therefore a

Box 39.3 Effects of Urinary Tract Obstruction on Renal Enzymes and Renal Gene Expression

Changes in Energy and Substrate Metabolism

Decreased oxygen consumption
Decreased substrate uptake
Increased anaerobic glycolysis
Decreased ATP/(ADP + AMP)
Decreased ammonia genesis

Changes in Enzyme Activity

Decreased
Alkaline phosphatase
Na⁺/K⁺-ATPase
Glucose-6-phosphatase
Succinate dehydrogenase
NADH/NADHP dehydrogenase
Increased
Glucose-6-phosphate dehydrogenase
Phosphogluconate dehydrogenase
Mitogen-activated protein kinases
Matrix metalloproteinase-2/-9
Mast cell protease 1 (chymase)

Changes in Gene Expression

Reduction in glomerular G_{as} and G_{αq/11} proteins
Reduction in preproepidermal growth factor and Tamm-Horsfall protein
Transient induction of growth factors FOS and MYC
Striking induction of cellular damage (*TRPM2*) genes
Induction of plasminogen activator gene

ADP, Adenosine diphosphate; *AMP*, adenosine monophosphate; *ATP*, adenosine triphosphate; *NADH*, nicotinamide adenine dinucleotide (reduced).

reduced driving force for lumen-to-basolateral magnesium flux across the paracellular pathway.²⁶⁰

TUBULOINTERSTITIAL FIBROSIS IN RESPONSE TO OBSTRUCTIVE NEPHROPATHY

An important focus of many experimental studies in obstructive nephropathy has been devoted to the effects of longer-term obstruction.^{157,261} In part, these studies use unilateral ureteral obstruction (UVO) as a convenient model for chronic kidney damage because the timing of the injury is clear and because the extent of injury should be reproducible from animal to animal.^{157,261} These studies, which have been conducted almost entirely in rodents, have elucidated an overall pathway for renal tubular epithelial damage and identified several potential targets for intervention. Obstruction inhibits oxidative metabolism and promotes anaerobic respiration, leading to decreased ATP levels and increased levels of adenosine diphosphate (ADP) and AMP.^{252,262,263} In addition, obstruction alters a wide variety of metabolic enzymes, as well as the expression of many different gene products.^{252,263-265} These changes are summarized in Box 39.3. Such changes in renal metabolism in UVO could contribute to the development

of renal fibrosis. In various studies, kidney tissue and fibroblasts showed increased expression of glycolytic enzymes including hexokinase [HK2], pyruvate kinase M2 [PKM2], and lactate dehydrogenase A [LDHA], along with higher lactic acid production during fibrosis.^{266,267} Notably, inhibiting PKM2 with shikonin and pyruvate dehydrogenase kinase-1 [PDK1] with dichloroacetate reduced fibronectin and collagen expression, as well as tubular apoptosis and macrophage infiltration in a 7-day UVO mouse model.²⁶⁸ Similarly, shikonin and 2-deoxyglucose, an aerobic glycolysis inhibitor, reduced myofibroblast activation in UVO-induced fibrosis.²⁶⁹ Many of these changes are difficult to link mechanistically with changes in GFR or tubular transport function observed in obstruction. It is possible, however, that reduced ability to generate ATP, along with reductions in Na⁺/K⁺-ATPase expression, contributes to the natriuresis observed after release of obstruction (see earlier discussion).

It is thought that chronic obstruction damages tubular epithelial cells by increasing hydrostatic pressure; reducing blood flow (due to the renal vasoconstriction that occurs in obstruction, see earlier discussion); and increasing oxidative stress.²⁶¹ In response, tubular epithelial cells release a number of autocrine factors and cytokines including angiotensin II,^{261,270} transforming growth factor-β (TGF-β),^{270,271} platelet activator inhibitor,²⁷² and tumor necrosis factor-α (TNF-α).²⁷³ These factors, along with the presence and increase in levels of adhesion factors, lead to the infiltration of the renal interstitium with inflammatory cells including macrophages. These, in turn, release additional cytokines. These factors accelerate the development of interstitial fibrosis by increased extracellular matrix, cell infiltration, apoptosis, and accumulation of activated myofibroblasts.²⁷⁴ In addition, there is upregulation of various receptors targeted by autocrine factors and cytokines including both type 1 and type 2 angiotensin receptors (AT1R and AT2R) and prostaglandin E2 receptors (EP2 and EP4).²⁷⁵⁻²⁷⁷ The complex cellular migration process may become amplified by epithelial-mesenchymal transition,²⁷⁸ which is characterized by downregulation of epithelial marker proteins such as E-cadherin, zonula occludens-1, and cytokeratin; loss of cell-to-cell adhesion; upregulation of mesenchymal markers including vimentin, α-smooth muscle actin, and fibroblast-specific protein-1; basement membrane degradation; and migration to the interstitial compartment.²⁷⁸ The entire cascade leads to tubulointerstitial fibrosis and permanent loss of kidney function, which may continue to progress after the obstruction has been relieved (Fig. 39.9).

It has been hypothesized that changes in the intratubular dynamic forces—so-called tubular stretch—in urinary tract obstruction also are an important determinant for development of tubulointerstitial fibrosis in the kidney.²⁷⁹ Thus both in vivo²⁷¹ and in vitro models^{242,280-282} of obstructive uropathy demonstrate that tubular stretch induces robust expression of TGF-β, proliferation, activation of tubular apoptosis, and induction of nuclear factor-κB signaling, which contribute to the inflammatory and fibrotic milieu.^{279,283} Because fibrosis is absent in mast cell-deficient mice with UVO, this suggests that mast cells play an important role in induction of the inflammation and development of tubulointerstitial fibrosis in obstruction.²⁸⁴ Furthermore, the importance of the renin-angiotensin

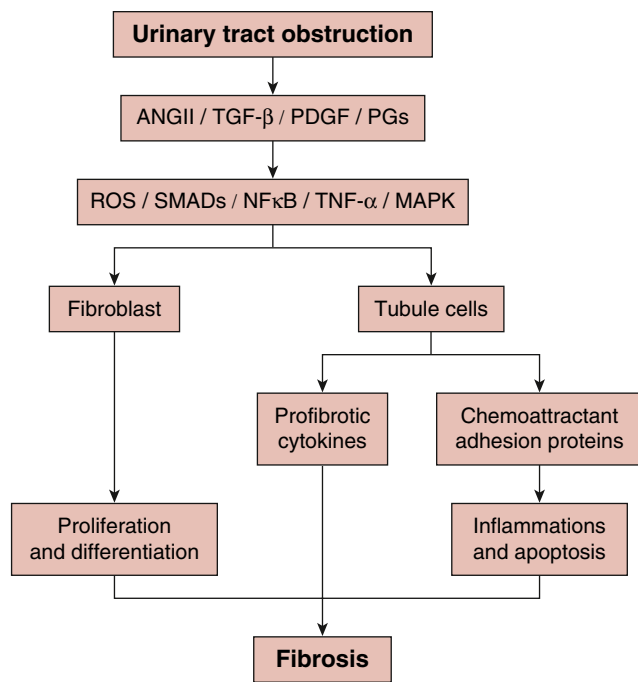


Fig. 39.9 Schematic overview of main signaling pathways involved in urinary obstruction-induced fibrogenesis. Binding of various factors to their receptors activates inflammatory and profibrotic signaling pathways, resulting in the formation and deposition of extracellular matrix (ECM) proteins. Prostaglandins (PGs) can be either profibrotic or antifibrotic depending on which receptor is activated. ANG II, angiotensin II; MAPK, mitogen-activated protein kinase; NF-κB, nuclear factor (NF)-κB; PDGF, platelet-derived growth factor; ROS, reactive oxygen species; TGF-β, transforming growth factor-β; TNF-α, tumor necrosis factor-α.

system for the development of fibrosis was also underscored by the observation that mast cells release renin possibly stimulated by autocrine histamine release.²⁸⁴ Interaction between DNA repair and parenchymal inflammation may also play a role for regulating fibrosis development during ureter obstruction. Apurinic/aprimidinic endonuclease (APE1) is vital for genome maintenance and transcription factor regulation. The main function of APE1 is repairing oxidative DNA base lesions caused by reactive oxygen species (ROS).²⁸⁵ Following 7 days of UUO, APE1 levels increased in almost all zones of the kidney and the transcription regulatory activity of APE1 is involved in the regulation of the profibrotic factor connective tissue growth factor.²⁸⁶ These findings suggest that APE1, with dual roles in DNA repair and transcription regulation, potentially plays a complex role in tubulointerstitial fibrosis development.

Thus the pathogenesis of renal fibrosis is a progressive and complicated process involving multiple molecular pathways and cellular targets including angiotensin II as highlighted already.^{4,175} Angiotensin II is known to stimulate the production of TGF-β, which is critical for the development tubulointerstitial fibrosis.²⁸⁷ Moreover, angiotensin II activates the NFκB signaling pathways, contributing to increased production of TNF-α, interleukin-6 (IL-6), and monocyte chemoattractant protein-1 (MCP-1), which promotes inflammation, apoptosis, and oxidative stress. Several studies have shown that antagonism of angiotensin II, TGF-β, TNF-α, or factors that attract inflammatory cells

may ameliorate UUO-induced damage.^{261,270-274,288-292} These studies propose that mast cells' release of renin and local angiotensin II formation in the obstructed kidney may lead to both vasoconstriction, which causes reduction in renal blood flow and GFR, and fibroblast and macrophage activation, which increases TGF-β levels and causes fibrosis of the kidney. Moreover, data suggest that mast cells also have the capacity to release chymase, a protease, which may limit development of tubulointerstitial fibrosis by decreasing infiltration of inflammatory cells and release of proinflammatory and profibrotic chemokines and cytokines.²⁹³

There is some controversy regarding the role of COX-2 in renal injury secondary to ureteral obstruction. Inhibition of COX-2 prevented UUO-induced apoptosis and fibrosis,^{294,295} indicating that COX-2 has detrimental effects. In line with these findings, it has been shown that injection of COX-2 chitosan/siRNA (small interfering RNA) nanoparticles in mice subjected to 3-day UUO diminished the obstruction-induced renal inflammation, apoptosis, and fibrosis.^{296,297} These chitosan/siRNA nanoparticles were designed to silence COX-2 specifically in the macrophages, indicating that specific knockdown of COX-2 in macrophages demonstrates a novel strategy to prevent ureter obstruction-induced kidney damage. In contrast, Kamata and colleagues²⁹⁸ found that meloxicam, a selective COX-2 inhibitor, worsened 5-day UUO-induced fibrosis, suggesting COX-2-derived prostaglandins' protective role against kidney damage. Supporting this, our study revealed increased apoptosis and tubular damage in COX-2 KO mice during 3 and 7 days of UUO,²⁷⁷ highlighting COX-2's protective effect. This was further supported by studies focusing on the PGE2, EP2, and EP4 receptors. Studies have demonstrated that both EP2 and EP4 receptors were upregulated during 7-day UUO,^{276,277} and we have recently shown that stimulation of the EP2 receptor mitigates renal fibrogenesis in 7dUUO mice.²⁹⁹ In addition, Nakagawa and coworkers³⁰⁰ showed aggravated kidney injury in mice lacking the EP4 receptor in response to 7dUUO. These studies confirm that the activation of EP2 and EP4 receptors might protect against UUO-induced damage, indicating that these receptors may be involved in the cytoprotective effect of COX-2/PGE2 activation during UUO. Similarly, augmentation of expression of factors that favor epithelial growth and differentiation, such as hepatocyte growth factor,³⁰¹ insulin-like growth factor, or bone morphogenic protein-7 (BMP-7),²⁶¹ may also have a protective effect.

Several studies propose that the primary trigger for initiating pathophysiologic processes is elevated pressure in the renal pelvis, leading to increased parenchymal pressure and subsequent mechanical stress. This stress activates stretch and swelling-activated cation channels within focal adhesions of epithelial cells, prompting a subsequent influx of Ca²⁺.^{281,283} This causes oxidative stress and stimulates migration of macrophages to the obstructed kidney. It has been shown that activation of AT1R on macrophages in UUO is essential to suppress their release of the proinflammatory cytokine interleukin-1β (IL-1β).³⁰² This suggests that AT1R on macrophages plays a crucial role in protecting the kidney from fibrosis by limiting interleukin-1 (IL-1) receptor activation. This finding might guide the development of effective therapies that address the limitations of broad

angiotensin receptor blockade. Thus the process leading to kidney fibrosis is complex, with multiple processes regulating the cellular responses responsible for these pathophysiologic changes.

Given species differences and the fact that obstruction in humans is often partial, the animal models may not predict entirely the behavior of postobstructive kidneys in humans. However, if the studies are relevant to human obstructive nephropathy, they suggest that patients undergoing release of obstruction may benefit from therapies that block proapoptotic, proinflammatory, or profibrotic mediators, or from treatments that stimulate epithelial cell growth and differentiation.^{261,270-274,288-290}

Experimentally, protection from obstruction-induced detrimental effects on kidney function can also be achieved by NO supplementation, which can be achieved through ACE inhibition, increasing kinin and NO levels, or by stimulating endogenous NOS with L-arginine.³⁰³ L-arginine is a semiessential amino acid and a key source of NO generation via NOS. In mice, chronic unilateral ureteral obstruction reduces inducible NOS activity and iNOS knockout mice exhibit more apoptotic tubules in the obstructed kidney, underscoring the important role NO plays in protecting the cellular functions in the obstructed kidney.³⁰⁴ Dietary L-arginine supplementation diminishes 3-day UO-related kidney damage in rats, suggesting its potential as a pharmacologic option in obstructive nephropathy.²⁶¹ Furthermore, treatment with α -melanocyte-stimulating hormone (α -MSH), a potent antiinflammatory hormone, was found to mitigate several harmful effects of obstruction. This underscores the pivotal role of inflammation in initiating deterioration of kidney function in urinary tract obstruction.³⁰⁵ It has been demonstrated that recombinant human erythropoietin (rhEPO) treatment inhibits the progression of renal fibrosis in the obstructed kidney and attenuates the TGF- β -induced epithelial-mesenchymal transition, suggesting that renoprotective effects of rhEPO could be mediated, at least partly, by inhibition of TGF- β -induced epithelial-mesenchymal transition.³⁰⁶

Renal fibrosis involves myofibroblast accumulation in the interstitium of the kidney, leading to deposition of extracellular matrix and organ dysfunction. Epithelial-mesenchymal transition may play an important role in this process. Reactivating Snail1 (encoding snail family zinc finger 1, known as “Snail1”) in mouse renal cells is essential for fibrosis development. Snail1 drives partial epithelial-mesenchymal transition, indirectly promoting myofibroblast formation and fibrogenesis by signaling to the interstitium, leading to sustaining inflammation.³⁰⁷ Importantly, it was shown that Snail1-induced fibrosis can be reversed in vivo and that obstructive nephropathy can be therapeutically ameliorated by targeting Snail1 expression. These results reconcile conflicting data on the role of the epithelial-mesenchymal transition in renal fibrosis and provide avenues for the design of novel antifibrotic therapies.³⁰⁷ The immune infiltrate in mouse kidneys with fibrotic obstructive nephropathy has been characterized and demonstrated that lymphocytes and macrophages were the most abundant immune cells present in the fibrotic kidneys.³⁰⁸ Preventing tubular epithelial cells from undergoing epithelial-mesenchymal transition also decreased immune infiltration, suggesting that crosstalk between tubular epithelial cells and

immune cells is of fundamental importance in sustaining inflammation and fibrosis.

Several other therapeutic avenues for direct targeting of specific pathways involved in the fibrotic process of obstructive nephropathy have been demonstrated. Apoptosis signal-regulating kinase 1 (ASK1) acts as an upstream regulator for the activation of p38 mitogen-activated protein kinase (MAPK) and c-Jun N-terminal kinase in kidney disease. Mice lacking the *Ask1* gene are healthy with normal homeostatic functions and are protected from renal interstitial fibrosis induced by ureteric obstruction.³⁰⁹ Thus the future may show therapeutic potential for ASK1 inhibitors to treat renal interstitial fibrosis.

FETAL URINARY TRACT OBSTRUCTION

Obstructive uropathy comprises the largest fraction of identifiable causes of impaired kidney function and kidney failure in infants and children. Compared with adult obstructive nephropathy, fetal obstructive nephropathy is particularly devastating because kidney growth and continued nephron development are impaired by the progression of fibrosis. Several studies have examined aspects of obstructive nephropathy in the newborn using a neonatal rat model of unilateral obstruction. The pathophysiology involved in fetal urinary tract obstruction is discussed in Chapter 71. Briefly, fetal urinary obstruction may lead to changes in tissue differentiation. At the time of birth, the rodent kidney is not fully developed and is representative of human renal development at about the midtrimester. Animal models reveal that fetal obstruction causes aberrations of morphogenesis, gene expression, cell turnover, and urine composition.^{310,311} The earlier the kidney is obstructed in utero, the greater will be the changes in kidney tissue.^{310,311} After birth, obstruction may affect kidney growth, especially in neonates and during the first year of life, but the obstruction will not cause tissue dedifferentiation.

TREATMENT OF URINARY TRACT OBSTRUCTION AND RECOVERY OF RENAL FUNCTION

Once the presence of obstruction is established, intervention is usually strongly indicated to relieve it. The type of intervention depends on the location of the obstruction, its degree, and its etiology, as well as the presence or absence of concomitant diseases, complications, and general condition of the patient.³¹² As detailed later, treatment options for obstruction include a wide spectrum of approaches, from active surveillance or minimally invasive endourologic techniques to open, laparoscopic, or robotic procedures. The main goal of therapy is to relieve symptoms and maintain or improve renal function. Initial emphasis should focus on prompt relief of the obstruction, followed by planning and hence executing the definitive treatment of its cause. Infravesical obstruction (e.g., benign prostatic hyperplasia, prostate cancer, and urethral stricture) is initially easily relieved with placement of a transurethral catheter. If the urethra is impassable, a suprapubic catheter may be needed. For obstruction above the bladder, insertion of a nephrostomy tube or ureteral stent may be

indicated. The urgency of the intervention depends on the degree of impaired kidney function, presence or absence of infection, and overall risk of the procedure.³¹⁰ The presence of infection in an obstructed urinary tract, or urosepsis, represents a urologic emergency that requires immediate relief of the obstruction, in addition to antibiotic treatment. Acute kidney injury, associated with bilateral ureteral obstruction or with the obstruction of single functioning kidney, also calls for urgent intervention.

Urinary tract stones constitute the most common form of acute unilateral urinary obstruction and can usually be managed conservatively with analgesics for control of pain and intravenous fluids to increase urine flow. Up to 75% of stones <6 mm pass spontaneously, but as calculi get larger, spontaneous passage becomes progressively less probable.³¹³ Active efforts to remove the stone are indicated for persistent obstruction, uncontrollable pain, or urinary tract infection. Current possibilities for treatment include semirigid ureteroscopy (usually with laser lithotripsy) for ureteral stones, flexible ureteroscopy for lower calyceal stones, shock wave lithotripsy for stones <2 cm, and percutaneous nephrolithotomy for stones >2 cm.³¹⁴ In the past, complex stones high up in the ureter or in the renal pelvis were difficult to remove without open surgery. However, improved methods of lithotripsy including the use of lasers through the ureteroscope, have made more stones amenable to fragmentation, while miniaturization of flexible ureteroscopes has made the entire upper urinary tract accessible in nearly all patients, except those with severe anatomic abnormalities.^{155,156} Once the stone has been removed, of course, appropriate medical (metabolic) therapy is needed to prevent recurrence.³¹⁵

Intramural or extrinsic ureteral obstruction may be relieved by placement of a ureteral stent through the cystoscope.³¹⁶ If this cannot be accomplished or is ineffective (especially in cases of extrinsic ureteral compression by tumor), then ultrasound-guided nephrostomy tubes will need to be inserted to effect prompt relief of the obstruction.³¹⁶

For infravesical obstruction due to benign prostatic hyperplasia, surgery can safely be delayed or completely avoided in patients with minimal symptoms, lack of infection, and an anatomically normal upper urinary tract.³¹⁷ If needed, transurethral resection of the prostate, laser ablation, or other techniques can be used for definitive treatment. Internal urethrotomy with direct visualization may be effective in the treatment of urethral strictures because dilatation usually has only a temporary effect. In recalcitrant cases, urethroplasty may be needed.³¹⁸

Patients with neurogenic bladder require medical management with anticholinergic drugs to decrease bladder pressures and prevent detrusor overactivity. This should be combined with institution of clean intermittent catheterization to enhance bladder emptying and decrease the risk of urinary tract infection. In circumstances where the aforementioned measures are unsuccessful and especially if the upper tracts become dilated, intradetrusor botulinum toxin A injections can be given, and ultimately bladder augmentation with use of an ileal segment might become necessary.^{33,319}

Robotic-assisted laparoscopic surgery has during the past 2 decades evolved from simple extirpative surgery to complex reconstructions including pyeloplasty for

hydronephrosis, ureteral reimplantation for obstructed megaureter, and a multitude of procedures for management of bladder, prostate, and gynecologic cancers.³²⁰⁻³²⁴ Studies comparing robotic-assisted laparoscopic procedures with their open surgical counterparts have generally shown surgical and oncologic outcomes to be equivalent but with lower analgesic requirements; estimated surgical blood loss and hospital lengths of stay were in favor of the robotic approach. The robotic approach is, however, more expensive.³²¹⁻³²⁴ A detailed discussion of the indications and surgical techniques for intervention to treat urinary tract obstruction is beyond the scope of this chapter and may be found in other sources.^{312,321,325}

ESTIMATING RENAL DAMAGE AND POTENTIAL FOR RECOVERY

Given the potential for meaningful recovery of kidney function, identification and correction of urinary tract obstruction requires vigilance and a swift response. In most cases, obstruction may be partial, so it is difficult based on history alone to predict the outcome. In addition, imaging studies that reveal both anatomy and function of the obstructed kidney predict the extent of functional recovery poorly (see earlier discussion) because the extent of anatomic distortion during obstruction correlates poorly with the extent of recovery once the obstruction is relieved.³²⁶ Renography with a variety of isotopes can be used to examine renal function, as outlined earlier. This approach is a far more reliable indicator of potential renal function when applied well after temporary drainage of the obstructed kidney (e.g., by nephrostomy tubes) has been achieved than if it is performed while the obstruction is still present.³²⁶ Imaging of the anatomy will provide information on the size and volume of the kidney but does not demonstrate reliable information on kidney function. Thus whether attempt procedures aimed at salvaging kidney function remains a challenging decision.

RECOVERY OF RENAL FUNCTION AFTER PROLONGED OBSTRUCTION

In patients, the potential for recovery depends primarily on the extent and duration of the obstruction. However, other factors, such as the presence of other conditions, including CKD and the presence or absence of urinary tract infection, play important roles as well. Recovery of kidney function in humans has been documented following release of obstruction of 69 days or longer.^{327,328} Because it is difficult to predict whether kidney function will recover when temporary relief of obstruction has been achieved, it makes sense to measure function repeatedly with isotopic renography over time, before deciding on a definitive surgical course. Chronic bilateral obstruction, as seen in benign prostatic hyperplasia, can cause kidney failure, especially when the obstruction is of prolonged duration and when it is accompanied by urinary tract infections.^{328,329} Progressive loss of kidney function can be slowed or halted by relieving the obstruction and treating the infection.

When obstruction has been relieved and there is a poor return of kidney function, interstitial fibrosis and inflammation may have supervened. To ensure that

there is no other process hampering recovery of kidney function, kidney biopsy may be indicated. As noted earlier, studies in experimental animals have implicated a variety of factors contributing to kidney failure due to prolonged obstruction, including excessive production of renal vasoconstrictors such as renin and angiotensin II and growth factors that may enhance fibrosis. On the basis of these findings, ACE inhibitors like captopril,³¹⁶ angiotensin receptor blockers,³³⁰ NO supplementation,³⁰³ α -melanocyte-stimulating hormone (α -MSH),³⁰⁵ and EPO³⁰⁶ have been shown experimentally to ameliorate to some degree the long-term damage to kidneys observed following prolonged obstruction.

POSTOBSTRUCTIVE DIURESIS

Release of obstruction can lead to marked natriuresis and diuresis with the wasting of potassium, phosphate, and divalent cations. It is notable that clinically significant postobstructive diuresis usually occurs only in the setting of prior bilateral obstruction or unilateral obstruction of a solitary functioning kidney. The mechanisms involved have been described in detail earlier and involve the combination of intrinsic damage to tubular cells, leading to impaired sodium, solute, and water reabsorption, as well as the effects of volume expansion, solute (e.g., urea) accumulation, and attendant increases in natriuretic substances such as atrial natriuretic peptide. When the obstruction is unilateral and there is a functioning contralateral kidney, the volume expansion, solute accumulation, and increases in natriuretic substances do not occur, and the contralateral kidney may retain salt and water, resulting in some compensation for the natriuresis and diuresis occurring in the postobstructive kidney. Management of the patient with postobstructive diuresis focuses on avoiding severe volume depletion due to salt wasting and other electrolyte imbalances, such as hypokalemia, hyponatremia, hypernatremia, and hypomagnesemia.

Postobstructive diuresis is usually self-limited. It typically lasts for several days to a week but may, in rare cases, persist for months. Acute massive polyuria or prolonged postobstructive diuresis may deplete the patient of Na^+ , K^+ , Cl^- , HCO_3^- , and water, as well as divalent cations and phosphate. Volume or free water replacement is appropriate only when the salt and water losses result in volume depletion or a disturbance of osmolality. In many cases, excessive volume or fluid replacement prolongs diuresis and natriuresis. Because the initial urine is isosthenuric, with an initial Na^+ of approximately 80 mEq/L, an appropriate starting fluid for replacement may be 0.45% saline, given at a rate somewhat slower than that of the urine output. During this period, meticulous monitoring of vital signs, volume status, urine output, serum and urine chemistry, and osmolality is imperative. This will determine the need for ongoing replacement of sodium, free water, and other electrolytes. However, some cases, especially in infants with bilateral obstructive nephropathy, have been described where the distal tubule remains refractory to aldosterone (pseudohypoaldosteronism) and patients develop paradoxical hyperkalemia despite a returning kidney function and a falling plasma creatinine.³³¹ In such cases,

during postobstructive diuresis, plasma potassium must be monitored carefully.

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QUESTIONS

1. How is hydronephrosis defined?
 - a. An obstructed kidney with impaired drainage of urine
 - b. A kidney with an enlarged renal pelvis
 - c. An enlarged kidney
 - d. A kidney with reduced size and parenchymal scars
 - e. An enlarged kidney with the presence of kidney stones

Answer: b

Rationale: It is consistent with anatomic changes.

2. How is urinary tract obstruction defined?
 - a. A condition with impaired kidney function
 - b. A condition with impaired kidney function due to vesicoureteral reflux
 - c. A condition where the surgeon finds the kidney obstructed
 - d. A condition where impaired urine drainage progressively leads to reduced kidney function
 - e. A condition that on computed tomography shows a hydronephrotic kidney

Answer: d

Rationale: It is consistent with functional changes in kidney function.

3. Which imaging studies are appropriate for diagnosis and investigation of hydronephrosis in an adult patient?
 - a. Ultrasound for detection of hydronephrosis
 - b. Computed tomography to identify a cause for obstruction
 - c. Renography to assess differential renal function
 - d. Diuretic renography to distinguish obstruction from stasis
 - e. All the above

Answer: e

Rationale: It is consistent with the added value of complementary examinations.

4. Why are renal blood flow and glomerular filtration rate impaired in response to urinary tract obstruction?
 - a. Because the increased pelvic pressure in the enlarged obstructed renal pelvis mechanically compresses the renal artery
 - b. Because urine drainage is reduced, and the obstructed kidney does not require a normal blood supply
 - c. Because glomerular filtration pressure is increased in the obstructed kidney due to constriction of the efferent glomerular arteriole
 - d. Because obstruction leads to vasoconstriction of the afferent glomerular arteriole
 - e. Because obstruction leads to vasodilatation of the afferent glomerular arteriole

Answer: d

Rationale: It is consistent with the well-characterized experimental evidence.

5. What are the pathophysiologic mechanism/mechanisms responsible for obstruction-induced tubulointerstitial fibrosis?
 - a. Release by tubule cells of multiple proinflammatory and profibrotic molecules including tumor necrosis factor- α (TNF- α), transforming growth factor- β (TGF- β), cytokines, and angiotensin II
 - b. Mast cell activation
 - c. Infiltration of the renal interstitium by inflammatory cells
 - d. Accumulation of interstitial myofibroblasts
 - e. All the above

Answer: e

Rationale: It is consistent with the complexity of the condition.